



**When the Decision Is Right but the Map Is Wrong:
Adversarial Robustness of Identity-Document Fraud
Detection and Localisation**

by

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Abstract

Identity-document fraud detection is challenging because manipulations can range from small textual changes, such as a single digit in a date of birth, to complex multi-region alterations involving portraits and identity-related text fields. Spatial outputs are used alongside forgery decisions to indicate where manipulation may have occurred. However, a correct document-level decision does not establish that the accompanying spatial evidence identifies the manipulated region. This dissertation investigates whether, under a white-box threat model, bounded adversarial perturbations can degrade initially acceptable spatial evidence while preserving the correct forgery decision. Such an attack can undermine trust in the spatial evidence accompanying an otherwise correct forgery decision. Two contrasting spatial-output pipelines are evaluated on identity-document imagery: ResNet-18 with Grad-CAM as a post-hoc attribution approach, and TruFor as a native manipulation-localisation approach.

The evaluation is organised around a three-stage DC–DCEC–DCEW framework. Decision Correct (DC) identifies manipulated images classified correctly before attack. Decision Correct, Evidence Correct (DCEC) retains the subset whose clean spatial maps satisfy joint relevance mass accuracy (E) and relevance rank accuracy (RRA) requirements, with thresholds calibrated using clean development evidence and visual review. Decision Correct, Evidence Wrong (DCEW) then records attacks that preserve the correct decision while causing either evidence requirement to fail. The study also conducts clean-data audits to examine possible shortcut cues and regional sensitivity, while classification-targeted attacks provide complementary controls. Each pipeline is assessed against its own clean baseline, with uncertainty estimated through document-stem resampling and shared eligible images used as an additional comparison view.

The study provides a detailed analysis of the FantasyID dataset and its limitations, demonstrates the value of complementary mass- and rank-based measures for assessing spatial evidence, and proposes the DC–DCEC–DCEW framework for evaluating the robustness of detection and localisation pipelines. It also highlights the need for future datasets that better represent the diversity and complexity of real-world identity-document fraud scenarios.

The study therefore evaluates a distinct reliability risk: spatial forensic evidence can require separate robustness assessment even when the associated forgery decision remains

correct. The conclusions are bounded by the selected dataset, model pipelines, spatial criteria and attack configurations, and do not constitute evidence of a verification bypass or of human-review performance.

Declaration

I have read and understood the University rules concerning academic integrity, plagiarism and appropriate referencing. I declare that the work presented in this dissertation is my own except where the contribution of others has been clearly acknowledged and referenced.

No substantial part of this dissertation has been submitted by me for assessment on another course or degree, except where this has been explicitly declared in accordance with University requirements.

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Chapter 1

Introduction

1.1 From physical to image-based verification

Modern identity documents have a long history, including medieval safe-conducts protected by the English Safe Conducts Act 1414 (McHugh, 2021; Parliament of England, 1414). Their value as evidence nevertheless depends on establishing that the document is genuine and belongs to the person presenting it.

The digital-first economy has changed how these documents are checked. Consumers now open bank accounts remotely, and the physical ID may be flattened into a smartphone photograph and examined through software. Remote services can inspect visible security features or incorporate chip reading, but a single photograph cannot provide the same evidence as handling a document or accessing its electronically stored information (Home Office, 2018). NIST’s guidelines recognise remote unattended proofing, where automated checks do not require interaction with a proofing agent (Temoshok et al., 2025a).

This workflow creates different opportunities for cyber criminals. Where an attacker can submit or inject a modified image, the forgery can target the captured pixels without requiring a convincing physical counterfeit. International security and law-enforcement agencies recognise identity and document fraud as facilitating wider criminal activity (Europol, 2025; Frontex, 2025). Driven by this changing threat landscape, researchers therefore evaluate both manipulation detection and spatial localisation (Luo et al., 2023; Guan et al., 2022) through shared datasets and benchmark competitions organised by bodies such as IAPR and NIST. The first asks whether a document was altered; the second asks where.

Spatial outputs also appear in commercial services. IDVerse reports using a neural network for document authentication, while Shufti provides heatmaps highlighting suspected alterations (LexisNexis Risk Solutions, 2025; Nikolett H., 2025). Such maps can support inspection, but neither commercial adoption nor greater spatial detail establishes their correctness.

1.2 From performance to dependable evidence

Problem addressed: A correct forgery decision does not establish that the accompanying spatial map identifies the altered region. This thesis investigates whether bounded adversarial perturbations can degrade initially acceptable spatial evidence while preserving that correct decision.

Neural forensic detectors learn patterns in image acquisition, editing and compression (Verdoliva, 2020). TruFor, for example, combines noise-sensitive features with RGB information to produce a spatial anomaly map and a document-level score (Guillaro et al., 2023). These outputs answer different questions and require separate evaluation.

Learned detectors may also exploit dataset-specific shortcuts. Compression or capture-device differences can separate training classes without representing the manipulation itself (Geirhos et al., 2020; Lapuschkin et al., 2019). Processing inconsistencies can provide useful forensic evidence, but they may also reflect how the dataset was assembled. The clean-data audit examines this distinction before adversarial degradation is interpreted.

Adversarial perturbations are deliberately optimised pixel modifications that target model behaviour within a specified budget (Goodfellow et al., 2015; Szegedy et al., 2014). It is important to consider that a document may already contain a swapped portrait or altered text before a subsequent perturbation is applied. Throughout this study, *clean* means before this additional perturbation; a clean image can therefore already contain a forgery. The concern is that a reviewer receives the correct warning with forgery detection but a less useful indication of where the alteration lies from spatial maps.

Figure 1.1 follows these routes from the document to the detector’s outputs. The left-hand branch shows how dataset shortcuts can undermine decisions or spatial evidence before an attack. The right-hand branch separates attacks, that change the decision from those that degrade spatial evidence while preserving it. The highlighted route with double red boxes is the focus of this thesis. The other routes explain the need for clean-data checks and classification-attack controls.

Forgery and detector-targeted perturbation must be analysed separately.

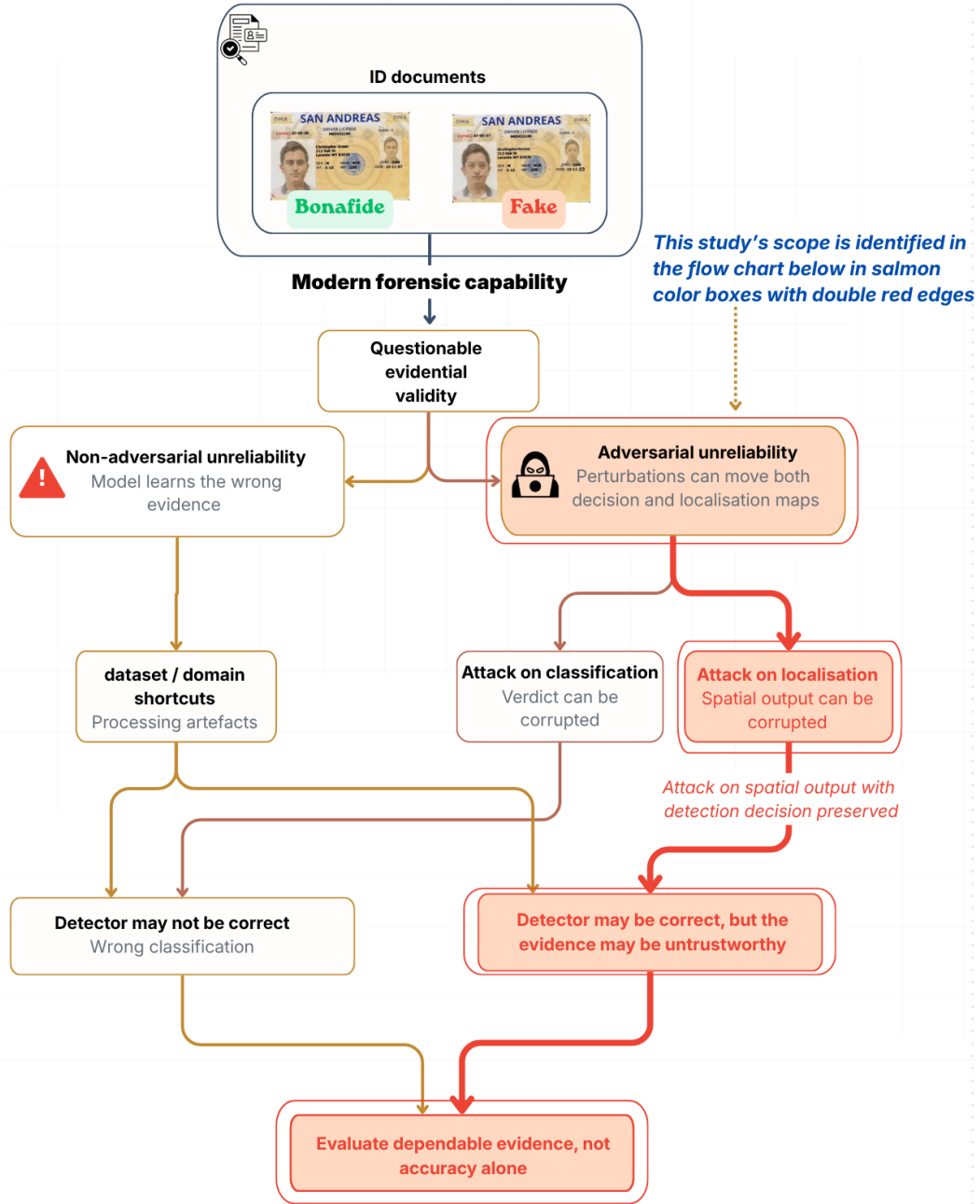


Figure 1.1: Conceptual routes to unreliable forensic outputs. Arrows show possible routes to failure; the highlighted route with double red boxes concerns degradation of spatial evidence while the correct forgery decision is preserved.

1.3 Two ways of producing spatial evidence, and what is not known

Learned detectors produce spatial evidence in two different ways.

1. **Post-hoc attribution:** A classifier returns a score, and an attribution method such as Grad-CAM is applied afterwards to indicate regions contributing to the target score (Selvaraju et al., 2017). The map is an inference about the classifier, rather than an output it was trained to produce.
2. **Native manipulation-localisation:** A purpose-built system such as TruFor is trained to estimate manipulated regions directly. Its spatial map is a supervised prediction, accompanied by confidence and detection outputs (Guillaro et al., 2023).

These outputs have different meanings. Agreement with manipulation annotations asks whether a map identifies the labelled regions. Explanation faithfulness asks whether it reflects the classifier’s behaviour. This study concerns the first property and its vulnerability to attack; spatial agreement alone does not establish the second.

Earlier work demonstrates prediction-preserving attacks on explanations and attacks on forensic localisers (Ghorbani et al., 2019; Dombrowski et al., 2019; Cao et al., 2024). The studies reviewed in Chapter 2 do not establish how initially acceptable spatial evidence degrades under a decision-preserving constraint in both kinds of pipeline on identity-document images.

FantasyID supplies complete document images, physical print-and-capture conditions, manipulated face and text regions, and public annotations (Korshunov et al., 2025a). ResNet-18 with Grad-CAM and TruFor represent the two routes. Each pipeline is evaluated against its own clean performance. The aim is to measure vulnerability within that performance, rather than to develop a higher-scoring detector.

The analysis follows three stages:

1. **DC(Decision Correct):** Identify manipulated images correctly classified before adversarial perturbation.
2. **DCEC(Decision Correct, Evidence Correct):** Within DC, retain images whose clean maps meet both a relevance-mass requirement and a relevance-rank-overlap requirement against the annotations.
3. **DCEW(Decision Correct, Evidence Wrong):** Within the fixed clean DCEC set, count attacks that preserve the correct decision but cause either evidence requirement to fail.

Relevance mass accuracy (E) measures how much map mass lies inside the annotation. Relevance rank accuracy (RRA) measures how far a set of the strongest map pixels, equal in size to the annotated region, overlaps it. “Evidence correct” therefore means passing this study’s spatial criterion. The protocol requires thresholds to be calibrated using clean development evidence and fixed before evaluating DCEC-to-DCEW outcomes. Both pipelines follow the same calibration procedure, although their numerical thresholds may differ. Within each pipeline, the same thresholds apply to clean and adversarial maps.

Reporting the numbers reaching DC and DCEC keeps limited coverage visible. Classification flips remain in the DCEC denominator and are reported separately. Images qualifying for both pipelines provide an additional view of each pipeline’s degradation; they do not remove differences in architecture, training or map resolution.

1.4 Aim, research question and objectives

This dissertation evaluates whether bounded adversarial perturbations can undermine initially acceptable spatial evidence while preserving a correct identity-document forgery decision.

Research question. To what extent can bounded adversarial perturbations degrade initially acceptable spatial evidence from post-hoc attribution and native localisation while preserving the correct identity-document forgery decision?

This question is investigated using ResNet-18 with Grad-CAM and TruFor. Each pipeline is evaluated against its own clean performance. Chapter 2 explains the reasons for selecting these models.

The following five objectives address this question.

- O1.** Establish clean detection and spatial-agreement baselines under documented preprocessing, reporting the images and forgery families each pipeline can support.
- O2.** Audit compression-related cues and regional sensitivity, and calibrate the joint E and RRA criterion using clean-development measurements and a visual review conducted without access to those measurements.
- O3.** Evaluate bounded, white-box attacks on spatial evidence, measuring DCEC-to-DCEW transitions and continuous degradation while reporting decision preservation. Use classification-targeted attacks as complementary controls.
- O4.** Assess each pipeline against its own clean baseline, quantify uncertainty using document-stem resampling, and examine shared eligible images and visual examples as an additional analytical view.

- O5.** Synthesise the findings to identify when spatial outputs support manipulation assessment, explaining the practical value, remaining risks and limits of the evidence.

1.5 Contribution

The contribution is an evaluation of attacks that undermine spatial evidence while preserving correct forgery decisions. Its outputs are clean coverage estimates, conditional DCEC-to-DCEW rates and continuous changes in spatial agreement. The clean audit supplies the context needed to distinguish an attack-induced failure from evidence that was already inadequate.

For financial-services practitioners, the study examines whether a map accompanying a correct forgery warning can become less useful for inspecting the document. It does not demonstrate a verification bypass or measure how human reviewers respond. For researchers, it provides a documented evaluation framework and experimental evidence for studying this failure.

The claims remain bounded by the selected dataset, pipelines and attack configurations. Rectangular annotations approximate manipulated regions, and the maps differ in meaning and resolution. These limits remain relevant when interpreting degradation relative to each pipeline’s clean baseline.

1.6 Structure of the dissertation

Chapter 2 reviews the datasets, spatial outputs, evaluation measures and adversarial literature. Chapter 3 defines the methodology, calibration and risk controls. Chapter 4 reports clean performance and DCEC coverage; Chapter 5 evaluates adversarial degradation. Chapter 6 discusses the findings and concludes. Supporting detail is organised in Appendix A for metric calibration and sensitivity checks, Appendices B and C for ResNet-18 and TruFor decisions and risk registers, and Appendix D for reproducibility records.

Chapter 2

Literature Review

2.1 Introduction

A forgery detector neural network pipeline can make the correct decision between bona-fide and forged identity document (ID) while giving a poor indication on the spatial map, as to where the alteration lies. This distinction is particularly important because IDs have many semantically distinct regions such as portraits, biographical text and document identifiers, coexisting with graphical and security features that are issuing authority specific. ID forgery may affect anything from a small part of a textual field to an entire portrait.

Before considering attacks on spatial evidence, it is therefore necessary to establish the characteristics of ID forgery as a forensic problem. This chapter reviews the literature needed to investigate whether an adversary can degrade an initially useful spatial map while preserving the detector’s forgery decision. It considers the domain-specific characteristics of manipulated IDs and the datasets available for studying them.

It then reviews two routes to spatial evidence followed by methods for evaluating spatial agreement and adversarial work targeting spatial outputs. These strands motivate the use of relevance mass and relevance rank accuracy and support this study’s DC–DCEC–DCEW framework.

The review connects findings to the research choices they support.

1. Dataset and pipeline selection inform the clean baselines in O1
2. Limitations of spatial measures inform the audit and calibration in O2
3. Adversarial studies inform O3 and O4.

Its scope is therefore a focused critical review of the literature needed to examine whether spatial evidence that is acceptable on a correctly detected forged ID can be degraded without changing the forgery decision. The research gap is this domain-specific conditional failure evaluation rather than a claim that spatial maps have not previously been attacked.

2.2 Identity-document forgery as a domain-specific forensic problem

International specifications for machine-readable travel documents define standards for physical dimensions, visual data layouts, and machine-readable zones to ensure global interoperability. The framework also addresses guidelines for protecting biographical data against alteration and establishes electronic chip and cryptographic standards (International Civil Aviation Organization, 2021). For certification of remote verification software, FIDO’s standardised test criteria include features such as holograms, guilloché patterns, microprinting, ghost images, specialised fonts, and interactions between the primary portrait and the document background (FIDO Alliance, 2026).

Consequently, IDs are a structured security artefact but not all of the physical security properties are observable from a single digital photograph used in remote verification. Some features depend on illumination, viewing angle, wavelength or electronic access. Since remote verification commonly operates on captured or uploaded document images, it makes it possible for alterations to the visually observable identity fields to reach an automated detector without the attacker constructing a complete physical counterfeit. The forensic problem considered in this thesis is restricted to this digital-image setting.

Within that setting, the spatial scale and semantic importance of a manipulation can vary substantially. Altering one or more characters in a date of birth, name or document number may modify only a very small image region while materially changing the represented identity. At the other extreme, a portrait may be partially modified or replaced entirely. The DeepID challenge identifies this combination of high-value targets including names, dates, identification numbers and facial portraits as a distinctive difficulty in ID manipulation detection, and treats document-level detection and pixel-level localisation as separate forensic tasks (Korshunov et al., 2025b). FantasyID models the same digital-manipulation setting using face-swapping and text-inpainting attacks and provides annotations of the manipulated regions (Korshunov et al., 2025a).

These properties also affect the learning formulation. A generic whole-image representation need not explicitly encode that a small change inside a date field can have greater forensic significance than a visually larger change in an irrelevant background region. Recent field-localised work therefore argues for explicitly modelling semantically important regions such as portraits and textual fields rather than treating the entire document as homogeneous image content (Kumar et al., 2026).

By explicitly representing manipulation classes during training, supervised modelling enables the system to learn the domain structure directly from the available labels. This does not mean that unsupervised or self-supervised approaches are unsuitable for ID fraud. A purely closed-set supervised detector may fail to represent new document layouts, fabrication processes or previously unseen fraud strategies. Recent work on open-set

ID-fraud discovery consequently combines document-specific representation learning with self-supervised and supervised objectives to identify anomalous and previously unseen fraud patterns (Li et al., 2026). Supervised detection and open-set anomaly discovery should therefore be viewed as complementary responses to different parts of the problem.

The present study concerns known digital manipulation families for which labels and alteration regions are available, making supervised detection and explicit spatial evaluation appropriate to the research question.

In remote verification pipelines, two naturally arising questions are:

1. Does the pipeline correctly classify the document as forged?
2. Does its spatial output direct evidence towards the known alteration?

A correct answer to the first question does not imply a useful answer to the second. A detector may exploit document-wide acquisition or processing traces, or other predictive cues outside the altered field, and still produce the correct document-level decision.

This separation between decision correctness and spatial agreement motivates the clean evidence conditions introduced later in this chapter. The adversarial perturbation studied in this thesis is a further, separate modification applied to an already forged image. Its purpose is not to create the underlying identity-document forgery, but to test whether initially acceptable spatial evidence can be degraded while the forgery decision is preserved.

2.3 Datasets, assurance criteria and the selection of FantasyID

Dataset selection is a trade-off between privacy, realism, scale, spatial ground truth and reproducibility as explained below. Dataset selection for this study’s objectives requires:

- R1: Complete document images.
- R2: Availability of underlying manipulation location.
- R3: Samples should retain acquisition characteristics relevant to remote verification.
- R4: Availability of clean and forged versions of the same image which would allow a bounded perturbation to the forged file without changing its manipulation label or altered region.

These requirements can be related to broader identity-proofing, AI-assurance and data-governance literature, although they do not prescribe a particular dataset.

Table 2.1 summarises how the principal frameworks which are a mix of legislations (the EU AI Act and GDPR), technical guidance (NIST), data-quality standards (ISO/IEC 5259) and adversarial-threat knowledge base (MITRE ATLAS) affect the criteria used in this study.

Table 2.1: External assurance frameworks and their implications for dataset selection and evaluation.

Framework	Relevant principle	Implication for this study
NIST SP 800-63A-4 (Temoshok et al., 2025b)	Identity evidence should be examined for authenticity, tampering, security features and correctness of core data. Automated document validation is an accepted validation mechanism.	Supports preserving whole-document context and security-relevant fields. A patch-only benchmark may contain useful forensic traces but cannot represent all evidence available to a document-validation pipeline.
NIST AI RMF 1.0 (Tabassi, 2023)	Accuracy and robustness should be assessed on realistic test sets representative of expected use, with relevant performance disaggregation and documented limitations.	Favours capture/device variation, manipulation-family reporting and explicit limits on external validity. It argues against treating performance on fictional document family as evidence of real-world generalisation.
EU AI Act, Arts. 10 and 15 (European Parliament and Council of the European Union, 2024)	For applicable high-risk AI systems, data governance addresses provenance, preparation, representativeness and data gaps; robustness requirements explicitly include AI-specific attacks such as adversarial examples.	Provides a useful assurance rationale for documenting dataset provenance and limitations and for adversarial robustness testing. It does not imply that the present research system is automatically classified as high-risk or legally compliant.
GDPR (European Parliament and Council of the European Union, 2016)	Purpose limitation, data minimisation and accountability constrain unnecessary processing and dissemination of personal data.	Explains the scarcity of unrestricted real-ID corpora and motivates fictional, synthetic or privacy-preserving datasets. Privacy protection can, however, conflict with the need to preserve full-document forensic context.
ISO/IEC 5259 series (International Organization for Standardization and International Electrotechnical Commission, 2024)	Data quality for analytics and machine learning includes fitness for purpose, quality measurement, management, labelling and lifecycle processes.	Supports examination of annotation quality, missing conditions, provenance and repeated observations rather than assuming that a large number of images alone implies a high-quality forensic benchmark.
MITRE ATLAS (The MITRE Corporation, 2026)	Catalogues adversarial tactics and techniques against AI-enabled systems, including adversarial-data crafting, model evasion and poisoning.	Primarily influences the threat model rather than dataset representativeness. A useful adversarial benchmark must permit controlled clean and attacked pairs and measurable preservation or degradation of model outputs.

Publicly available ID datasets make different compromises between capture realism, manipulation ground truth, privacy and reproducibility. Table 2.2 compares their fit to the requirements of this study; a limitation here is task-specific rather than a judgement of overall dataset quality.

Table 2.2: Critical comparison of identity-document datasets for whole-document spatial and adversarial evaluation.

Dataset	Whole document	Physical capture	Relevant digital forgery	Alteration ground truth	Privacy / realism trade-off	Critical role for this study
MIDV-2020 (Bulatov et al., 2022)	✓	✓	—	—	Mock documents enable reproducible capture variation without exposing real identities.	Strong capture-condition benchmark, but it does not provide the paired forgery regions required for spatial-evidence evaluation.
FakeIDet2-db (Muñoz-Haro et al., 2026b)	—	✓	~	Patch-level	Patches retain forensic texture from real IDs while reducing disclosure of personal information.	Useful real-document forensic evidence, but the released patch representation removes the whole-document coordinate system required here.
SIDTD (Boned et al., 2024)	✓	✓	✓	✓	Uses mock identities and MIDV-derived documents while adding recaptured forged examples.	Closest public alternative. Its crop-and-replace and inpainting construction creates a different manipulation setting and may introduce attack-specific boundary cues.
IDNet (Guan et al., 2024)	✓	—	✓	~	Large synthetic corpus provides scale and fraud diversity with low privacy exposure.	Useful for diversity and cross-layout analysis, but synthetic rendering does not reproduce the same physical print–capture history.

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Table 2.2 continued

Dataset	Whole document	Physical capture	Relevant digital forgery	Alteration ground truth	Privacy / realism trade-off	Critical role for this study
FantasyID (Korshunov et al., 2025a)	✓	✓	✓	✓	Fictional cards reduce privacy and legal constraints while retaining physical acquisition before manipulation.	Selected: combines whole images, physical capture, face/text digital manipulation and spatial annotations needed for the DC–DCEC–DCEW evaluation.
DeepID private set (Korshunov et al., 2025b)	✓	✓	✓	✓	Real identity documents improve ecological relevance but cannot be publicly reproduced.	Important external evidence for realistic field-level attacks, but unavailable as the reproducible experimental corpus.
FakeIDet3-DB (Muñoz-Haro et al., 2026a)	—	✓	✓	Patch-level	Privacy-aware contextual patches retain evidence from real and manipulated identity documents.	Closer to this thesis’s digital threat model than FakeIDet2, but still lacks the complete-document spatial context required by the primary evaluation.

Legend: ✓ = directly supports the criterion; ~ = partially supports it or supports it in a different form; — = not provided in the form required here. “Alteration ground truth” refers to spatial information sufficient to evaluate the known manipulated region. “Relevant digital forgery” refers principally to face and/or text manipulation rather than recognition-only or presentation-attack data.

No dataset dominates all of these criteria. A recurring trade-off is visible between *forensic realism* and *privacy-preserving reproducibility*.

FantasyID provides the closest conjunction of the required properties. Its fictional cards are physically printed and recaptured before digital manipulation, so the forged samples retain a physical acquisition history while face and text regions are subsequently altered using contemporary generative tools (Korshunov et al., 2025a). It also preserves the complete image and known alteration regions. These properties permit the same forged image to be assessed for the DC-DCEC-DCEW framework of this study.

These strengths come with two cautions from its authors’ later work. bona-fide and attacked images differ systematically in JPEG quality, a shortcut a detector can exploit without examining the edited field, which is why Chapter 3 controls compression before training. In the DeepID challenge, detectors that performed well on FantasyID’s text edits fell to chance on Poisson-blended edits of real IDs (Korshunov et al., 2025b). Because DeepID shares FantasyID’s authors and generators, this is the strongest available transfer warning rather than independent validation.

Three limitations or risks remain after dataset selection:

1. Rectangular annotations approximate the altered regions
2. Capture variants of one card are dependent observations
3. Fictional documents cannot establish performance across real issuing authorities.

These motivate mitigating strategies including mask-sensitivity checks, grouping by document stem and restricted claims. Chapter 3 defines these controls, Appendix A records metric-related risks and the model-specific registers retain preprocessing decisions.

2.4 Two routes to spatial evidence

2.4.1 Post-hoc attribution and ResNet-18 with Grad-CAM

Grad-CAM uses gradients of a target score to weight convolutional feature maps, producing a coarse class-discriminative localisation map (?). Because it is applied after classifier training, localisation supervision is not required. ResNet-18 provides an established residual classifier that remains tractable for repeated training and adversarial optimisation (He et al., 2016). The combination therefore provides a conventional classifier–attribution pipeline without making architecture development part of the research question.

Alternative attribution methods represent different design choices. Integrated Gradients is a path-based attribution method that explains a prediction relative to a reference input and satisfies Sensitivity and Implementation Invariance, although its result depends on the

baseline (Sundararajan et al., 2017). Score-CAM represents a forward-pass, perturbation-style CAM alternative that avoids gradient-based weighting but increases computational cost when explanations must be generated repeatedly (Wang et al., 2020). Finer-CAM addresses a different choice namely “the explanation target”, by contrasting a target class with a reference class to suppress shared features (?). None of these alternatives converts classifier attribution into a supervised forgery mask. Grad-CAM is therefore retained as an established and computationally suitable representative, not as a claim that it is the most faithful attribution method.

Finer-CAM does also show that contrasting a target class against a reference class can suppress features shared between classes and expose finer discriminative cues (?). This motivates a contrastive manipulated-versus-bona-fide target in the binary classifier, but does not establish that such a map will coincide with the document alteration. The exact target and layer are therefore methodological choices defined in Chapter 3 and Appendix B.

Spatial resolution imposes a separate limitation. Upsampling a coarse convolutional map cannot recover character-level information absent from its feature grid. More importantly, disagreement with the manipulation mask is not automatically evidence that Grad-CAM inaccurately represents the classifier: a correct decision may genuinely depend on compression or other document-wide cues. Clean spatial agreement must therefore be established separately before adversarial degradation is interpreted.

2.4.2 Native localisation and TruFor

TruFor approaches the problem differently. It combines RGB features with Noiseprint++, a learned representation of camera and processing traces, through transformer-based fusion (Guillaro et al., 2023). It produces an anomaly map and an associated confidence map; confidence-weighted pooling summarises these outputs before a classifier produces the image-level integrity score. Unlike Grad-CAM, localisation is therefore an explicitly trained component of the forensic pipeline. This structural connection makes decision-preserving spatial degradation a stronger constraint, while still leaving open whether substantial map changes can occur without changing the final decision.

Similar to post-hoc attribution, Native localisation also has alternatives offering different design choices. TruFor represents a general-purpose forensic model with explicit manipulation localisation, while DeepID demonstrates document-specific adaptation of pretrained forensic models (Korshunov et al., 2025b). EdgeDoc represents a more directly domain-adapted alternative by combining visual and forensic-trace features specifically for identity-document forgery detection and localisation (?). Such systems may benefit from task-specific training, but adopting them would also change the amount and type of domain supervision in the comparison. TruFor is therefore retained as a pretrained native-localisation baseline rather than as the strongest possible document-specific model.

A document-trained ResNet-18 and pretrained TruFor do not have equal domain exposure, but fine-tuning both would not equalise their objectives, pretraining or supervision either. Architecture-level ranking would consequently be misleading. Each pipeline is evaluated against its own clean evidence, while results on shared eligible images provide a controlled supplementary comparison rather than evidence that one architecture is intrinsically superior.

2.5 What should count as acceptable spatial evidence?

2.5.1 Spatial agreement is not explanation faithfulness

A spatial map can look convincing without reflecting what the model has learned. Randomisation tests show that some saliency methods retain substantial structure after learned parameters are disrupted (Adebayo et al., 2018). Such tests concern faithfulness to the model; comparison with a manipulation mask asks whether displayed evidence coincides with the known alteration. The two can diverge because a classifier may faithfully rely on compression, acquisition or other shortcut cues outside the edited field. Here, “acceptable spatial evidence” therefore means spatial agreement with the alteration, not a general certificate of explanation faithfulness.

Standard segmentation measures are also an awkward common criterion. Pixel-F1 and intersection-over-union suit binary regions, while TruFor reports F1 using both a fixed threshold and a best per-image threshold (Guillaro et al., 2023). Grad-CAM instead produces relative attribution strengths rather than calibrated manipulation probabilities. A common threshold would therefore impose different assumptions on the two pipelines. F1 and IoU remain useful diagnostics, but continuous-map measures avoid making binarisation the primary basis of comparison.

2.5.2 Why relevance mass and relevance rank are combined

Score-CAM considers the proportion of saliency energy inside an annotated region (Wang et al., 2020), while CLEVR-XAI formalises relevance mass accuracy (RMA), written E below and relevance rank accuracy (RRA) for non-negative maps (Arras et al., 2022). Let $S(p)$ be the spatial value at pixel p , M the annotated manipulation region, and $K = |M|$:

$$E = \frac{\sum_{p \in M} S(p)}{\sum_p S(p)}, \quad \text{RRA} = \frac{|\text{TopK}_K(S) \cap M|}{K}. \quad (2.1)$$

The measures test different properties.

- E asks how much total map mass lies inside the alteration
- RRA asks whether the strongest responses preferentially occupy it.

A map can perform well on one while remaining weak on the other, so their combination provides a stronger basis for judging clean spatial evidence than either alone.

Their transfer from CLEVR-XAI nevertheless needs qualification. Its relevant objects are known by construction, whereas an ID-document mask marks where an edit occurred, not necessarily every cue legitimately used by a detector. The map semantics also differ:

- Grad-CAM expresses class attribution
- TruFor produces learned manipulation scores.

The measures therefore provide a common test of spatial agreement, not equivalent explanations.

2.5.3 Limitations of spatial agreement metrics

No localisation metric is independent of its evaluation setting as discussed below.

1. CLEVR-XAI reports changes in relevance accuracy with target size, while broader work shows that saliency metrics can disagree and that implementation choices can alter conclusions (Arras et al., 2022; Tomsett et al., 2020). This is important for identity documents, where a large portrait replacement and a small text edit differ greatly in area.
2. Simpler alternatives have complementary weaknesses. The Pointing Game asks only whether the maximum response lies inside the target (?); one successful peak cannot establish regional agreement.
3. Conversely, overlap measures require thresholding. Rectangular or union annotations can also conceal uneven localisation across components.

The literature therefore supports combining spatial properties rather than treating one score as a universal definition of evidence quality. These findings provide the conceptual basis for separating a correct decision (DC) from a correct decision with acceptable clean spatial evidence (DCEC). The operational definition of DCEC, threshold calibration, supporting diagnostics and the later decision-preserving adversarial failure condition are specified in Chapter 3.

2.6 Attacking spatial evidence while preserving forgery decisions

Ghorbani et al. (2019) show that small input perturbations can alter attribution maps while preserving the predicted label. Dombrowski et al. (2019) steer explanations towards

chosen targets and relate the effect to network geometry. Together they establish the separation used here: prediction stability does not imply spatial-evidence stability. Their experiments concern generic image classification and mainly measure departure from the clean explanation.

In the controlled benchmark setting used here, dataset annotations identify the true manipulated region, which may range from a few characters to an entire portrait. This allows spatial degradation to be measured against the actual edit while the forgery decision is preserved, even though such ground truth is unavailable in deployment.

Heo et al. (2019) instead modify model parameters to mislead explanations while retaining predictive performance. This is a model-owner threat, distinct from a bounded input-space adversary. Audio-deepfake work extends prediction-preserving explanation attacks to a forensic task, but uses domain-specific perceptual constraints and map-similarity objectives (Kitłowski et al., 2026).

Within the identity-document literature reviewed here, work has focused on detection and localisation rather than deliberately degrading initially acceptable spatial evidence while preserving the forgery decision (Korshunov et al., 2025a,b; Kumar et al., 2026). This motivates testing both post-hoc attribution and native localisation, measuring degradation against the known manipulation rather than map change alone, and conditioning adversarial failure on acceptable clean evidence.

2.7 Attacking forensic and document localisers

Cao et al. (2024) demonstrate optimisation- and gradient-based attacks on image-tampering localisers, including transfer between models. Zeng and Pun (2024) introduce frequency-constrained attacks to improve transferability. Their results establish vulnerabilities in spatial forensic prediction, but neither supplies this study’s joint test of acceptable clean evidence followed by its loss under a preserved image-level decision.

Document-specific adversarial robustness has also begun to receive attention. Shao et al. (2024) evaluate attacks on DocTamper and T-Sroie and propose manifold-based adversarial training. Their experiments show localisation degradation and improvements from defensive training. This is a closer domain precedent than natural-image attacks: sparse tampered pixels and weak forensic features affect both the attack and its evaluation. However, their segmentation-focused study does not evaluate Grad-CAM and TruFor through a clean DCEC population with a preserved document-level forgery verdict. The remaining gap is consequently narrower than adversarial robustness of documents in general.

Projected-gradient methods provide a way to search within a declared perturbation budget (Madry et al., 2018). It should be noted that a budget only bounds pixel changes and does not by itself prove imperceptibility or realistic access to a deployed service.

Evaluation guidance also warns that weak attacks and unusable gradients can produce misleading robustness claims (?). White-box evaluation is therefore a controlled stress test. Successful attacks establish vulnerability under the tested conditions; unsuccessful attempts do not certify robustness. This evaluation literature therefore motivates explicit attack-strength and gradient-validity checks in the methodology, without treating successful localisation degradation as evidence of a practical verification bypass.

2.8 Synthesis and research gap

The reviewed literature establishes the main components of the problem separately.

- ID studies distinguish forgery detection from localisation and include manipulations ranging from small text edits to large portrait changes (Korshunov et al., 2025a,b).
- Explanation research shows that attribution maps can change while the predicted class remains stable (Ghorbani et al., 2019; Dombrowski et al., 2019).
- Forensic-localisation studies show that manipulation maps are themselves vulnerable to adversarial attack (Cao et al., 2024; Shao et al., 2024).
- Spatial-evaluation work further shows that different metrics capture different properties of continuous maps (Wang et al., 2020; Arras et al., 2022).

What remains less directly studied is their intersection in identity-document forensics: whether a forged document that is both correctly detected and initially supported by acceptable spatial evidence can have that evidence degraded by bounded perturbations while the forgery decision remains correct. This distinction is important because changing an already poor map does not demonstrate loss of useful evidence, while changing the document decision is ordinary evasion rather than the failure considered here.

The research gap is therefore narrower than adversarial robustness or localisation failure in general. It concerns the extent to which bounded perturbations degrade initially acceptable spatial evidence from post-hoc attribution and native localisation while preserving the correct identity-document forgery decision. DC, DCEC and DCEW provide the conceptual states needed to express that distinction; their operational definitions belong to Chapter 3. The objectives in Chapter 1 follow this sequence from clean capability and evidence qualification to adversarial degradation, comparison and interpretation.

2.9 Conclusion

The literature supports treating identity-document forgery as a structured forensic problem in which correct detection and useful spatial evidence are distinct requirements. FantasyID

Table 2.3: Literature synthesis relative to the research question.

Literature strand	Established	Remaining question
ID-document forensics (Korshunov et al., 2025a,b)	Detection and localisation are distinct; manipulations vary greatly in scale.	Can initially useful spatial evidence be selectively degraded while the forgery verdict remains correct?
Explanation attacks (Ghorbani et al., 2019; Dombrowski et al., 2019)	Predictions may remain stable while attribution changes.	Does that change correspond to loss of agreement with the true document alteration?
Forensic-localiser attacks (Cao et al., 2024; Shao et al., 2024)	Manipulation-localisation outputs are adversarially vulnerable.	Does decision-preserving evidence loss also occur in a forgery-detection pipeline?
Spatial evaluation (Wang et al., 2020; Arras et al., 2022)	Mass and rank capture complementary spatial properties.	How should loss of initially acceptable evidence be distinguished from change in an already inadequate map?

provides annotated face and text manipulations, while ResNet-18 with Grad-CAM and TruFor represent post-hoc and native routes to spatial evidence. Existing explanation and localisation attacks establish relevant vulnerabilities, but do not directly resolve the decision-preserving loss of initially acceptable evidence posed by the research question.

Chapter 3 translates this literature-derived rationale into the experimental protocol; Chapters 4 and 5 report the clean and adversarial evidence.

Chapter 3

Methodology

3.1 The question and the shape of the study

A correct forgery warning does not tell us whether the map alongside it points to the alteration. This study asks how far a small, deliberately chosen change to a document image can reduce initially acceptable spatial evidence while keeping that warning correct. Chapter 2 explains why the two outputs need to be measured separately. This chapter explains how the images, maps and attacks are prepared and how a failure is counted. Chapters 4 and 5 present the clean and adversarial findings.

The investigation uses two pipelines that produce different kinds of map. ResNet-18 is a document classifier; Grad-CAM is applied after its prediction to attribute the forged-class score to parts of the image (He et al., 2016; ?). TruFor is a pretrained forensic system with a native manipulation map and a separate document score (Guillaro et al., 2023). The purpose is to test each pipeline’s evidence against its *own* acceptable clean evidence. Different training exposure, map meanings and input resolutions prevent a difference between their raw scores from being attributed to architecture alone.

The unit of the main analysis is a manipulated document image for one pipeline, paired with that pipeline’s adversarial version of the *same* image. We first identify clean-correct forged decisions (DC). Among these, we identify clean maps that meet a joint spatial-evidence rule (DCEC). We then examine the already attacked versions of that fixed DCEC set: a preserved correct decision with evidence falling below either part of the rule becomes DCEW. Showing the original number of forgeries, DC coverage and DCEC coverage is essential. Otherwise a high conditional attack rate could describe a tiny fraction of the documents that originally entered the detector. Figure 3.1 depicts the denominator trail; its clean-evidence gate is still being calibrated at this revision.

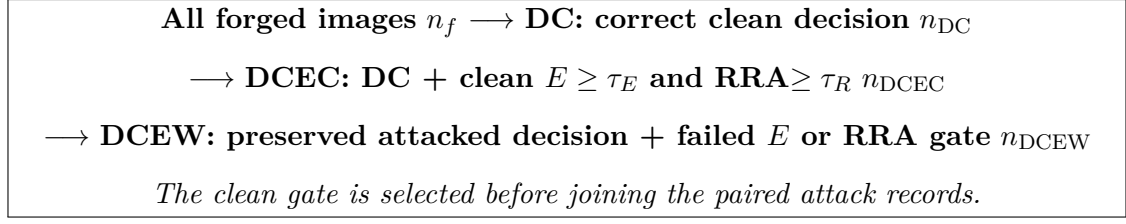


Figure 3.1: Analysis populations for one pipeline. The clean DCEC denominator is fixed before attacked outcomes are counted. This analytical order does not imply the attacks were run after the gate was designed; calibration and counts remain pending.

The methods contribute to Objective O1 through the clean detection and map audit, O2 through the compression check and clean-evidence calibration, O3 through constrained attacks and classification controls, and O4 through paired and shared-image analysis. Objective O5 is addressed by the synthesis in Chapter 6. The regional restoration and compression probes help rule out simplistic accounts of a model’s behaviour; they are supporting diagnostics rather than extra measures of attack success.

3.2 What the measurements can establish

The study takes an empirical, fallibilist view. A model score, the pixels of a map and the result of applying a fixed perturbation are measurable properties of these particular computational pipelines. That is the practical *ontological* assumption: the outputs exist whether or not they match an assessor’s interpretation. The *epistemological* limit is that “good evidence” is not observed directly. It is inferred from agreement with annotated rectangles, a visual judgement and an explicitly chosen rule. A rectangle may cover an edited field without tracing its exact boundary; a classifier might use a real editing trace just outside it. Agreement therefore supports a bounded claim about the map and annotation, not proof that the model reasons like a forensic examiner.

This leads to a mainly quantitative, within-image experimental design. Holding the pipeline and document fixed makes clean-versus-attacked change meaningful; family-wise results show where the effect is concentrated. A small structured author review of clean maps provides context for choosing an evidence threshold, but it does not measure whether a human analyst would detect fraud more accurately. The work combines deductive tests of a stated decision-preserving failure with exploratory diagnostics prompted by observed clean behaviour. Those post-hoc checks are labelled as such, and their results are not treated as independent confirmation of a hypothesis formed before viewing the data.

3.3 Documents, splits and preparation

FantasyID was chosen because it supplies complete photographed cards, altered face and text regions, and manipulation annotations (Korshunov et al., 2025a). Forgeries are already present in the images before our adversarial changes. In this dissertation, *clean* means the image as received by the chosen pipeline before that additional change; it does not mean an unaltered identity document. No live customers or submitted real identity documents are recruited for the model evaluation.

The open 1,899-image FantasyID training release was divided by card stem into a project-training set of 1,440 images from 160 stems and an internal development set of 459 images from 51 *different* stems. The latter contains 306 forged images (153 Digital-1 and 153 Digital-2) and 153 bona-fide images. The separately restricted FantasyID validation release was not used. The official test has 1,385 images: 786 Digital-3, 150 FaceDancer, 149 TextDiffuserFT-BFEI and 300 bona fide. All available images enter the clean detection summaries; spatial summaries are conditional on correct clean forgery decisions, with the original family counts retained. This gives good coverage of this benchmark, but the numbers of cards and manipulation types still restrict generalisation.

Stem separation prevents related captures of a card from appearing on both sides of the *internal* training–development split. It does not make the whole official test card-independent. An audit of Digital-3 found that 480 of its 786 test edits have a matching card/capture parent in project training and 153 have one in project development; 153 do not have a matching project parent. The *manipulated test images and the Digital-3 manipulation family* were not used to fit the ResNet model, but it would be inaccurate to call that family a wholly unseen-card test. The effects of shared parents, devices and the official test’s different bona-fide pool are considered in the interpretation. No unsupported identity-disjoint claim is made for the other official-test families.

The dataset provides altered-region rectangles. Within each image, we clip each rectangle to the valid raster and use the union of the resulting pixels as the reference region. We record empty or unusual boxes for review. In particular, 108 TruFor-native rectangle clips await case-level checking. The reference region is an approximate annotation, not a pixel-perfect segmentation or a guarantee that every useful forensic cue is inside the box. Bona-fide cards have no altered-region target: they enter classification specificity and TruFor’s *decision-threshold* fitting, but not the forged-image *E/RRA* gate fitting.

3.3.1 A common compression condition

Early clean results could have reflected how dataset files were prepared. A logistic-regression probe using JPEG-history features alone separated project-development classes with AUROC 1.000 after fitting on project-training features. This made compression a plausible shortcut even if the model also noticed actual alterations (Geirhos et al., 2020).

We therefore use the documented Policy-C condition for *both* final pipelines. The source JPEG is decoded to RGB; the pixels are encoded at JPEG quality 75 with 4:2:0 chroma sampling and decoded again; a second encoding uses a deterministic quality from 50 to 90 with the same sampling. That final quality is shared by images with the same card stem and capture source. Figure 3.2 shows where this operation sits before the two pipeline-specific input transforms.

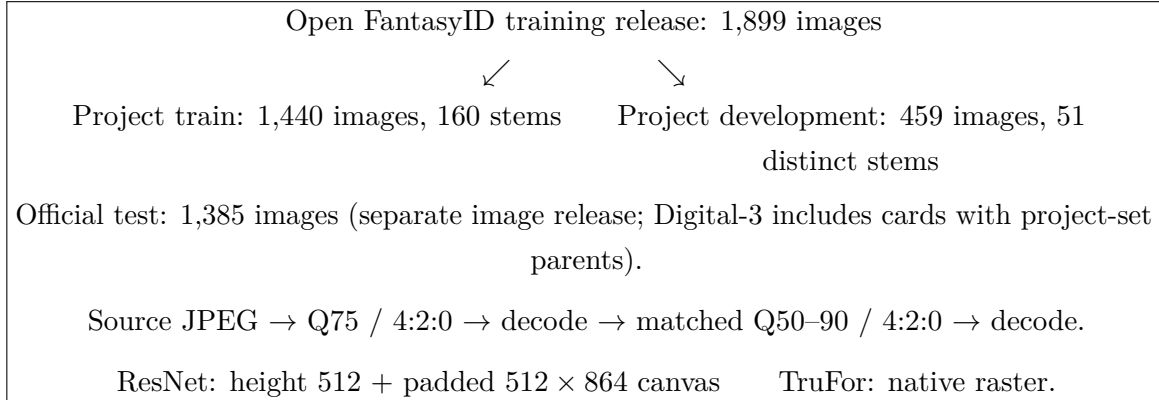


Figure 3.2: Internal stem split, official test and common Policy-C preparation before pipeline-specific transforms. The official test is not asserted to be card-identity-disjoint.

The probe is repeated on the controlled images in Chapter 4. Direct final-JPEG table features then have development AUROC .500, while a broader combination still reaches .586. We reduce an easy class cue without claiming to remove every processing trace. Recompression can also remove legitimate forensic information. The results therefore describe these prepared inputs; they are not a test of the original source JPEGs or a deployed submission workflow. Appendix C records the alternatives and residual risk.

3.4 The two evaluated pipelines

3.4.1 ResNet-18 and Grad-CAM

ResNet-18 begins with ImageNet-1K pretrained weights and is fitted to the project-training condition. Training first updates its classifier head for three epochs with the feature extractor frozen, then updates the whole model for twelve epochs. The selected full-model checkpoint comes from epoch 12 by development AUROC. One recorded training seed was used. These choices make a runnable baseline and avoid training a full document detector from scratch on only 160 card stems; one seed does not measure variability across model fits. The same development data inform checkpoint selection and its reported development score, so that score is partly apparent rather than independent validation.

These stages are one sequential transfer-learning schedule. Keeping the pretrained feature extractor frozen and fitting only a new head treats its features as fixed; unfreezing

all layers allows the features themselves to adapt to photographed documents. The three head-only epochs warm up the new classifier before the twelve full-model epochs. This study evaluates the resulting selected model. It does not present frozen-only training, direct full-model training and the sequential schedule as three controlled comparison arms.

The Policy-C document is resized to height 512 while keeping its aspect ratio, then centred on a 512×864 canvas. Padded sides are excluded from spatial analysis and stay unchanged during adversarial optimisation. The actual document RGB is ImageNet-normalised before the network. Its forged decision uses a probability cut-off of 0.5. Grad-CAM uses the final residual block and the forged-minus-bona-fide logit; it weights that block’s activations by their spatially averaged gradients, retains positive relevance and interpolates the resulting map back to the document canvas (?). The coarse underlying map and the resizing matter when a text alteration is only a few pixels wide. Grad-CAM shows attribution for the chosen score; it is not a separately supervised edit segmentation.

3.4.2 Pretrained TruFor

The final TruFor condition retains the released pretrained weights and accepts the full Policy-C RGB image at native resolution (Guillaro et al., 2023). The manipulated-class softmax output is the spatial map. Its pooled image-level detection score, which also uses the model’s confidence information, is a *different* output. The final forged-score threshold, 0.532956, was chosen for pooled image accuracy on all 459 project-development images and then applied unchanged to every family and capture source. A calibrated decision threshold does not turn the weights into a fine-tuned model; “zero-shot” refers here to the absence of FantasyID weight adaptation. Calibration and reporting on the same development images produce an apparent score, and the single threshold behaves unevenly across capture sources.

An earlier TruFor fine-tuning experiment used a different split and preparation protocol. It is kept as a diagnostic in Appendix D, with no results spliced into the final pretrained condition. The pipelines consequently differ in training exposure as well as architecture and map type. This makes *within-pipeline paired change* the primary inference; shared images offer a secondary descriptive check.

3.5 From a map to clean evidence eligibility

For a forged image i , let M_i be its non-empty altered-region union, Ω_i its valid document pixels and $S_{im}(p) \geq 0$ the map for pipeline m on those pixels. These symbols refer to the *relevant pipeline’s raster*: the same edit can have different pixel area after ResNet resizing

and on TruFor’s native image. Define $K_i = |M_i|$ on that raster. For a finite map with positive total mass, the two main spatial measures are

$$E_{im} = \frac{\sum_{p \in M_i} S_{im}(p)}{\sum_{p \in \Omega_i} S_{im}(p)}, \quad \text{RRA}_{im} = \frac{|\text{TopK}_{K_i}(S_{im}|\Omega_i) \cap M_i|}{K_i}. \quad (3.1)$$

E describes the share of all map mass that falls in the annotation. RRA asks what share of the strongest K_i pixels lies there, with K_i equal to the annotation’s area. This use of mass and rank adapts relevance measures to the document setting (Arras et al., 2022); the exact combined pass rule is this study’s choice. A map with strong evidence on one edited face but weak evidence on a second, smaller text field can still score well on the union. For context we also report annotation fraction $A_i = |M_i|/|\Omega_i|$, area-normalised concentration $\mu_{w,i} = E_{im}/A_i$, the Pointing Game indicator for the strongest pixel, and face/text components where annotations permit. None of these supporting measures replaces the joint eligibility rule.

The computation uses the document’s valid coordinates: ResNet’s artificial padding is excluded, and TruFor’s full native content is retained. The same mask and coordinate convention must be used for the clean and attacked version of an image. Map interpolation and tied values at the top- K boundary can change RRA; Appendix B states the tie and invalid-map checks. A zero-mass, non-finite or empty-mask case is identified separately, rather than quietly treated as evidence correct. The TruFor RRA implementation and common-coordinate audit remain to be completed before the final evidence gate is frozen.

Let $d_{im}(x)$ mean that pipeline m correctly calls the forged image x manipulated at its *fixed* decision threshold. After a pipeline-specific pair of clean-evidence thresholds has been selected, its analysis sets are

$$\text{DC}_m = \{i : d_{im}(x_i) = 1\}, \quad (3.2)$$

$$\text{DCEC}_m = \{i \in \text{DC}_m : E_{im}(x_i) \geq \tau_{E,m}, \text{RRA}_{im}(x_i) \geq \tau_{R,m}\}, \quad (3.3)$$

$$\text{DCEW}_m = \{i \in \text{DCEC}_m : d_{im}(x_i^{\text{adv}}) = 1, [E_{im}(x_i^{\text{adv}}) < \tau_{E,m} \vee \text{RRA}_{im}(x_i^{\text{adv}}) < \tau_{R,m}]\}. \quad (3.4)$$

The attack cannot enlarge its denominator by adding poorly localised clean images. A flipped decision remains in the original DCEC denominator and is reported as a *separate* outcome. Undefined attacked maps also require a declared counting convention and an alternate tally; that decision has not yet been frozen. These explicit categories prevent a missing map from being passed off as either a successful localisation or an observed continuous E value.

3.5.1 Choosing a defensible evidence rule

The literature supplies the E and RRA measures, but no universal threshold for a photographed identity-document rectangle. The intended reference is a structured, masked visual review of clean development maps. For images with face and text edits, the author rates each field on an ordinal 0/1/2 scale and records uncertain cases and a short reason. During coding, the individual E , RRA, provisional pass labels and adversarial outcomes are hidden. Images are shown in mixed order with their annotated edit and card context. The author has already seen some aggregate experimental findings, so this is *not* prospectively independent or fully blinded human validation. The coding sheet, panel presentation, review dates and any repeat ratings must be retained.

Once the ratings are locked, the planned calibration compares candidate E /RRA pairs with a declared visual reference, using balanced accuracy to avoid rewarding a trivial majority call. Both pipelines follow the same procedure; their numeric thresholds may differ because their maps mean different things. Candidate grid, reference-label conversion, tie breaks and invalid-map policy must be written down before fitting. The clean development material alone supplies those choices. No threshold is adjusted against official-test maps, already observed attack losses or a preferred DCEW rate. The chronology matters: some DC attacks were completed *before* this joint gate was designed. Withholding per-image attack outcomes during the remaining coding limits direct tuning, but it does not erase the analyst’s earlier exposure. Appendix B gives the full review protocol and sensitivity checks.

[RESULT PENDING: finish the face/text review, verify TruFor RRA and coordinate geometry, record and freeze the selected $\tau_{E,m}$ and $\tau_{R,m}$ with their clean-only selection rule, then insert the DC→DCEC counts and coverage in Chapter 4. Do not report a DCEW rate until this is done.]

3.6 Adversarial changes and the comparison attack

The threat model gives the experimenter white-box access to each model, its gradients, the annotated altered-region union and the correct forgery label. It asks whether a feasible bounded image *can* move the spatial output; it is more informed than a person submitting a document to an unknown service. Following adversarial-evaluation practice, the objective, constraint, input domain and success check are all specified (Madry et al., 2018; ?). Both evidence-directed arms seek to decrease E on the already forged image while keeping that image on the forged side of its pipeline’s decision threshold. They do not optimise RRA directly; RRA must be measured afterwards as an independent part of the joint outcome.

For either pipeline, write z_0 for its Policy-C image in physical RGB units $[0, 1]$, after the ResNet resize when applicable. A proposed image z must satisfy $\|z - z_0\|_\infty \leq 1/255$

and the valid RGB range. Starting exactly at z_0 , the evidence attack makes ten projected sign-gradient proposals with step size $0.25/255$. If a proposal crosses the forged decision boundary, eight bisections towards the last feasible image try to recover a decision-preserving point. Only a feasible proposal with $E_{\text{proposed}} \leq E_{\text{current}} + 10^{-8}$ is retained; otherwise the previous feasible image is kept. There is one start and no restarts. The tolerance permits numerical ties and does not promise a strict decrease on every step. After optimisation, a separate fixed evaluator checks each final score, map and actual perturbation size. Appendix A supplies the pseudocode, coordinate conversions and implementation record; Figure 3.3 outlines the two model-specific paths.

ResNet-18 / Grad-CAM	Pretrained TruFor
Resized document RGB, padding fixed; differentiate E through forged-class Grad-CAM.	Native document RGB; differentiate E through manipulation map.
Forged logit margin ≥ 0 .	Separate forged score ≥ 0.532956 .
Both: $\ \delta\ _\infty \leq 1/255$ in physical RGB; 10 projected proposals, eight feasibility bisections when needed.	
RRA is calculated after the search, not optimised.	

Figure 3.3: The two evidence-directed searches have a common physical RGB budget but different input rasters, maps and decision checks.

ResNet’s search changes only the resized document-content pixels. ImageNet channel normalisation turns its physical RGB budget into channel-specific network-space bounds. Decreasing E requires differentiating through the gradients used to make Grad-CAM, so the search involves higher-order derivatives. Its hard feasibility rule is forged-minus-bona-fide logit ≥ 0 , equivalent to the 0.5 forged-probability decision here. This is an attack on the processed 512×864 input, not a claim that a perturbation to an original full-resolution JPEG would have the same effect after resizing.

TruFor may change any native document pixel, with no artificial padding. Its cached $[0, 1]$ image z is passed to the released model as $x = z \cdot 255/256$. Consequently the same physical $1/255$ bound becomes $1/256$ in these model coordinates, and the $0.25/255$ step becomes $0.25/256$. Feasibility requires its separate forged score to stay at or above 0.532956 . Full native gradients required recomputation of activations and exact attention-query chunks to fit available GPU memory. The gradient route was checked against ordinary evaluation where both fit; a separate unmodified model evaluates attack candidates. The single largest-image one-step test in Chapter 5 establishes implementation feasibility, not a full-population attack effect.

For both arms the authoritative attacked object remains continuous floating-point RGB. It is not converted to eight-bit pixels or passed through a new JPEG save before the

final reported map and decision. A $1/255$ norm bound is therefore a bounded *model-input* stress test, not a guarantee of imperceptibility, physical reproducibility or survival through a real upload pipeline. Model-specific spatial results must also be read alongside their different preprocessing and initial clean coverage.

A separate ResNet control targets the *bona-fide decision* with ten-step projected gradient descent at the same $1/255$ bound and $0.25/255$ step, but starts at a random point inside the allowed box. Three 72-image pilots tested $4/255$, $2/255$ and $1/255$; the latter is the smallest *tested* budget, not a proven minimum. The full control uses all 456 selected images across Digital-1, Digital-2 and FaceDancer, including six FaceDancer cases already classified incorrectly. Its outcome asks whether a classifier can be fooled, and helps explain why an attribution can vanish after the forged class vanishes. Such a flipped case cannot satisfy the decision-preserving DCEW definition. A comparable full TruFor classification control has not been verified at this revision.

3.7 Who enters each result and how change is estimated

The ResNet evidence attack was completed on 450 initially DC images: 153 Digital-1, 153 Digital-2 and 144 FaceDancer. The first two are internal development families; FaceDancer supplies the selected official-test attack family. ResNet DC elsewhere includes just 6 of 786 Digital-3 and 32 of 149 TextDiffuserFT forgeries, with very weak clean regional evidence. These are displayed in the original-family clean coverage rather than silently reclassified as attack failures or successes. The selected attack scope was chosen using clean performance, and any later DCEC count will be conditional on that scope. The frozen TruFor DC roster contains 1,330 images: 153 Digital-1, 135 Digital-2, 786 Digital-3, 107 FaceDancer and 149 TextDiffuserFT. Partial Phase-A records have been ingested and audited for scientific state, but the full paired ten-step population and its RRA outcomes still need validated final outputs; a processed count alone is not an attack outcome.

Chapter 4 reports pooled detection in the challenge style as well as the full family-by-family DC and map distributions. Overall classification performance includes genuine-card specificity; the forged-only DC denominator for E/RRA is stated separately. The distinct view in Chapter 5 first describes continuous clean-to-attacked change among paired DC cases, then, once the clean gate is frozen, estimates the conditional DCEC-to-DCEW fraction. For an image with positive clean E , its relative E loss is $(E_i^{\text{clean}} - E_i^{\text{adv}})/E_i^{\text{clean}}$. We average *those image-wise losses*; subtracting family means and dividing by the clean family mean would answer a different question. Positive and negative individual changes stay in the analysis.

For the final DCEC report, each pipeline and family should show the chain $n_{\text{forged}} \rightarrow n_{\text{DC}} \rightarrow n_{\text{DCEC}}$. Classify each eligible attack in order: incomplete run; changed decision (while recording any invalid map as a separate flag); preserved decision with invalid map; preserved decision with valid map and a failed evidence rule (DCEW); or preserved decision with valid map and a passing rule. These mutually exclusive bins keep the fixed eligible denominator intact. Alongside $n_{\text{DCEW}}/n_{\text{DCEC}}$, report $n_{\text{DCEC}}/n_{\text{forged}}$ and the absolute counts. A sensitivity analysis can count decision-preserved invalid maps as evidence failures, but must show that choice explicitly. The percentages then show both *susceptibility where a map started useful* and *how often that starting point existed*. A shared eligible-image intersection may show the *same documents* for each pipeline’s separate paired calculation, including paired visual examples; it does not support an unadjusted ranking of the architectures.

Images from the same card stem are related. The completed ResNet relative- E intervals resample document stems, keeping their images together, and recompute the paired statistic in each resample. The proposed DCEW intervals must likewise recompute both numerator and frozen clean DCEC denominator for resampled stems. A separate repeat of clean-only threshold fitting can examine calibration uncertainty; a stem interval conditional on a single chosen gate does not capture it. Nor does it capture retraining, the uncertain rectangle boundaries, alternative attack starts or a new institution’s documents. Family-wise summaries and a rule-based set of median, low-change, large-change and borderline examples help show exceptions without passing selected images off as a rate. Appendices B and E give the rating and example-selection details.

3.8 Validity, ethics and a traceable record

Several design choices address specific risks. Compression checks and the controlled JPEG condition address a preparation shortcut, although residual history remains. Stem separation protects the internal model-development split; the Digital-3 parent overlap is disclosed rather than treated as a wholly new identity test. Family-wise denominators expose poor clean detection. A second spatial measure, face/text inspection, annotation-boundary checks and visual review address the risk that a good union- E conceals a poor map. The classification control separates decision flips from evidence failures. Exact evaluator checks and gradient parity help avoid mistaking an engineering failure for adversarial robustness. The dates, measurements and open actions for these risks are summarised in Appendix A; model-specific registers remain in Appendices C, D and E.

The material is a research dataset of fictional documents; no external human participants were recruited for the model evaluation. The author-only visual review is a method for interpreting clean maps, not a study of customers or independent forensic experts. Images and example panels may nevertheless depict plausible personal details or identifiers.

Working copies, access controls and any public dissemination of examples must follow the dataset terms and applicable university data-storage rules. **[AUTHOR VERIFY BEFORE SUBMISSION: state the actual WMG ethics approval, delegated approval or waiver route; approval date/reference and whether author coding required a separate review; the actual storage location, access and retention plan; and cite or attach the approval/waiver form, confirmation and required training records. These have not been verified from the available experiment files.]**

The key reproducibility unit is a dated set of data-split rules, Policy-C settings, chosen model/decision threshold, clean evaluator, attack protocol and per-image paired results. Chapter prose states the decisions needed to understand the result; Appendix A holds the procedure and a provenance template for scripts, run manifests, checkpoint IDs and file digests. Those identifiers should be inserted only after checking the actual files. Hardware-dependent numerical differences, one ResNet training seed and the unexplained small difference between an earlier and the final ResNet visual attack run limit exact repetition. The final visual run supplies the Chapter 5 paired estimates; its corresponding individual records, rather than a mixture of runs, must be preserved.

3.9 Conclusion

The design begins with a real clean-baseline question: which documents get a correct warning, and which of their maps agree well enough with the edited region to be worth testing? It then measures what changes when a bounded, annotation-aware attack tries to reduce that agreement while the warning remains correct. The two pipelines share the analytical rules but keep their actual maps, input coordinates and clean populations separate. Chapter 4 establishes the observed coverage and clean spatial behaviour; Chapter 5 reports completed paired attacks and distinguishes their present limits from the DCEC-to-DCEW result still dependent on the frozen clean gate.

Chapter 4

Clean Performance and Spatial Evidence Audit

4.1 Why clean evidence matters

The question in this thesis is what happens to a useful spatial map when an adversary changes the image but leaves a correct forgery decision intact. Hence it is imperative to establish which clean decisions are correct and whether their maps point to the altered regions. “Clean” here means *before the additional adversarial perturbation*; the document may already contain an edited face or text field.

This chapter addresses following objectives.

- O1 by measuring document decisions and spatial agreement for both pipelines.
- O2 by checking a compression shortcut and testing which edited regions influence model’s decision.

It reports the clean DC (decision correct) population and the continuous map measures from which DCEC (decision correct, evidence correct) will be drawn. A final DCEC count is also provided using a joint evidence thresholds and visual review. This distinction keeps failures that occur *before* an attack visible.

The details of the spatial measures and review procedure are in Appendix B. ResNet and TruFor decisions, alternatives and residual risks are documented in Appendices C and D. Chapter 3 describes how the evaluated pipelines were built, including ResNet’s ImageNet initialisation, classifier-head training and subsequent full fine-tuning.

4.2 Images, splits and evaluation conditions

FantasyID supplies photographed documents and annotations of the altered regions (Korshunov et al., 2025a). It has a total of 3,284 images split into training(1,899) and

test(1,385) dataset. The open 1,899-image training release was divided by document stem into 1,440 images from 160 card-stems for this project’s training and 459 images from 51 different stems for project development. The public release dataset audit established the units, dependencies and family semantics required for valid interpretation of the experiments.”Figure 4.1 describes the process with discovered features about the dataset while Figure 4.2 shows auditory information derived. This *internal* development split must not be confused with the separately released, restricted FantasyID validation set, which was not used due to its EULA restrictions.

FantasyID Public Dataset Structure evaluation

Public release anatomy • manifest → derivation → reproducible card-level split

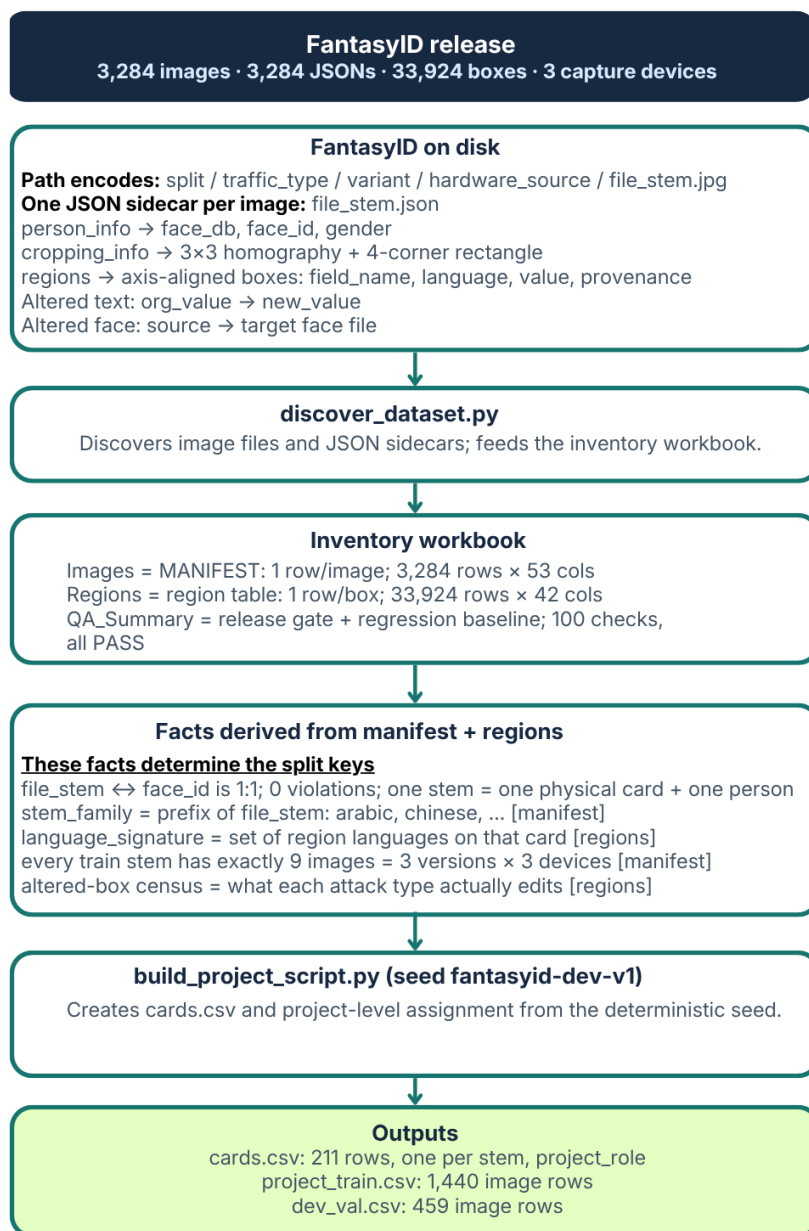


Figure 4.1: Evaluation process of Fantasy ID dataset along with discovered information.

Fantasy ID dataset analysis derived audit information

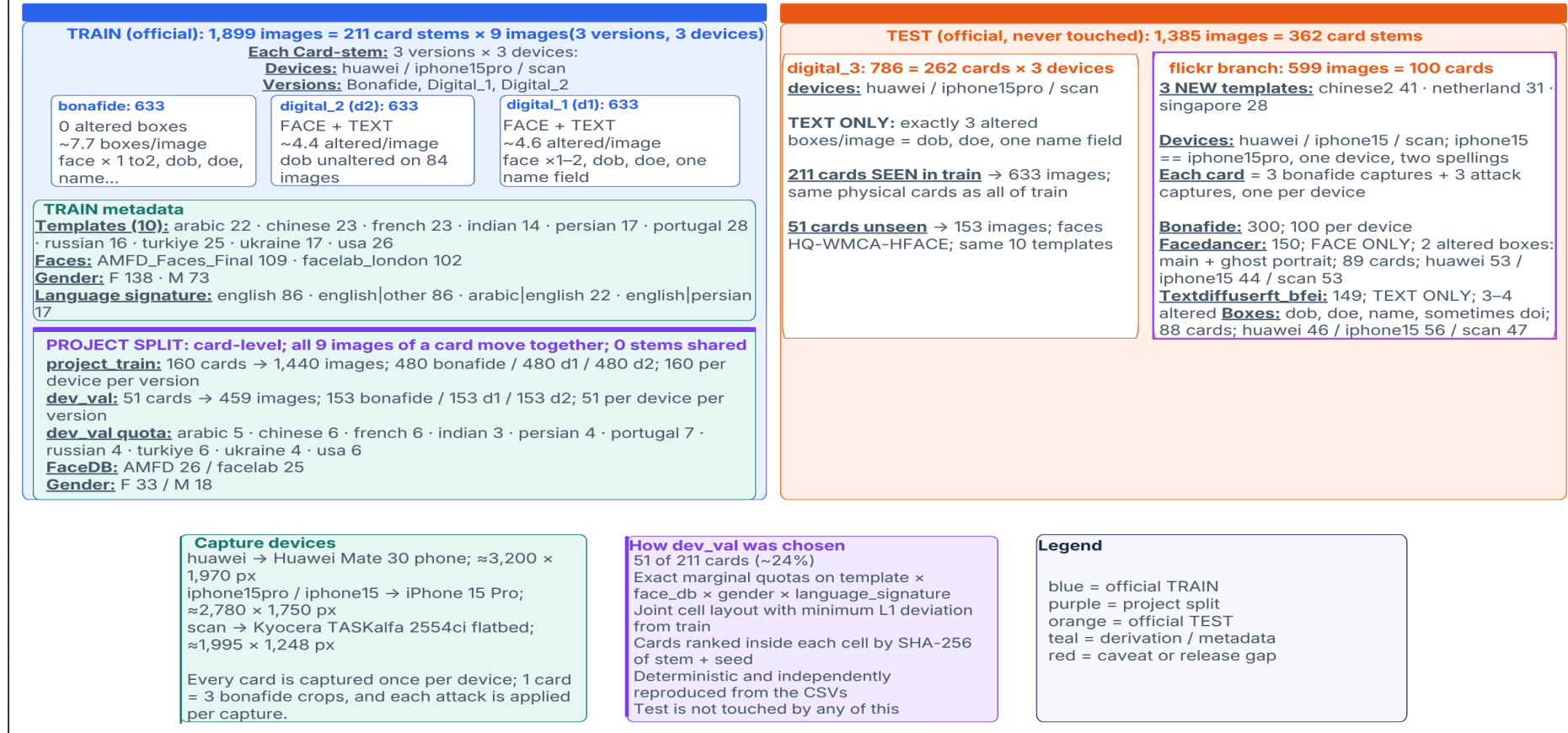


Figure 4.2: Fantasy ID dataset audit information.

ResNet-18 was trained on Digital-1 and Digital-2 categories and receives a resized document within a padded canvas. Grad-CAM is calculated from the forged-minus-bona-fide logit at the last residual block. TruFor keeps its pretrained weights, reads the document at its original resolution and produces a map of manipulation likelihood. Its *decision* threshold was selected using the project-development images; its weights were not adapted in the final condition. “Zero-shot” therefore describes the weights, not an absence of calibration.

The development results are partly apparent performance: they informed ResNet checkpoint selection and TruFor threshold calibration. The official test prompted some later diagnostics, which are identified below as post-hoc checks. Training exposure, image resolution, map meaning and calibration all differ between pipelines. The main question remains how each pipeline’s spatial output holds up against *its own* clean baseline.

4.3 Checking the compression shortcut

An initial audit found an alternative explanation for very good development detection. JPEG-history features alone separated forged from bona-fide development images with AUROC 1.000. The audit fitted a simple *logistic-regression classifier* on compression features from project training, then tested it on the stem-disjoint development set. It showed that preparation history could reveal the class without examining the edited face or field. This is the kind of shortcut discussed in Chapter 2 (Geirhos et al., 2020).

Policy C was adopted to reduce this class-correlated cue. Table 4.1 gives the actual image sequence; the two JPEG encodings are separate, with a decode between them. The final quality is deterministic for a document stem and capture source, so its forged and bona-fide versions receive the same assignment. Chapter 3 and Appendix C give the implementation and the reasons for choosing this policy.

Table 4.1: Policy C at a glance. The same final quality is assigned to images from the same card and capture source. This is an outline of the implemented sequence, not a new image enhancement method.

Step	Input and operation	Output
1	Decode the source JPEG to RGB.	Source pixels in memory.
2	Encode those pixels as JPEG at quality 75, 4:2:0.	Common intermediate JPEG.
3	Decode the intermediate JPEG to RGB.	Intermediate pixels in memory.
4	Encode at a deterministic, matched quality from 50 to 90, 4:2:0.	Generates the Policy-C image used in evaluation.

Using the same kind of logistic-regression probe after this control, following was observed.

- Final JPEG-table features alone gave development AUROC 0.500.
- Broader JPEG-history features gave 0.524.
- All measured compression features together gave 0.586, with a stem-resampled 95% interval [0.572, 0.606].

These are *different feature sets*, not three scores for one unchanged input. The aim was to harmonise a preparation cue correlated with the class, not to remove every encoding trace a real edit might leave.

Both evaluated pipelines use Policy-C images. Recompression could weaken a useful forensic trace, while residual histories could still influence a decision. The results describe this controlled condition, not performance on the source files or in a bank. The regional intervention below, tests whether ResNet also responds to edited content under the control. It cannot restore traces lost through recompression or exclude every shortcut. These limits are retained in the Appendix C risk register.

4.4 Document decisions across forgery families

DeepID competition style overall reporting provides a useful first view: it includes false alarms on genuine cards as well as missed forgeries. Table 4.2 uses class-weighted F1, as in the competition’s detection reporting, alongside accuracy, balanced accuracy and the underlying correct counts (Korshunov et al., 2025b). These are *this study’s* Policy-C results at its chosen decision thresholds. They are not challenge leaderboard scores: the inputs and TruFor threshold differ from the challenge protocol, and the private competition set was not evaluated here. DeepID also scores binary localisation masks; the spatial measures later in this chapter answer a different question.

Table 4.2: Pooled clean detection under this study’s protocol. “Forged DC(Decision-Correct)” and “BF(Bona-fide) correct” give the numerators and denominators behind the summary. Accuracy, balanced accuracy (BA), class-weighted F1 and AUROC are calculated separately for each split.

Split	Pipeline	Forged DC	BF correct	Accuracy	BA	Weighted F1	AUROC
Dev	ResNet-18	306/306	148/153	.989	.984	.989	1.000
Dev	TruFor	288/306	73/153	.786	.709	.769	.843
Test	ResNet-18	182/1,085	209/300	.282	.432	.274	.234
Test	TruFor	1,042/1,085	131/300	.847	.699	.831	.944

An overall figure as noted in Table 4.2 and used in competitions cannot show *which* kind of forgery a model detects or whether its spatial map points to the edit. Table 4.3

therefore retains the family results. AUROC is the chance that a randomly selected forgery receives a higher forgery score than a randomly selected genuine document, across decision thresholds. Each family AUROC uses that family’s forgeries and the bona-fide pool from the *same split*. DC and balanced accuracy refer instead to the selected operating threshold. Balanced accuracy is (forged recall + bona-fide specificity)/2; the same genuine pool is reused for each family in a split.

Table 4.3: Clean detection by forgery family. Forged n is the original family size. DC is the number correctly called forged; BA includes genuine images from the same split. All results use the final fixed decision thresholds.

			ResNet-18			TruFor		
Split	Family	Forged n	AUROC	DC/ n	BA	AUROC	DC/ n	BA
Dev	Digital-1	153	1.000	153/153	.984	.986	153/153	.739
Dev	Digital-2	153	1.000	153/153	.984	.699	135/153	.680
Test	Digital-3	786	.059	6/786	.352	.999	786/786	.718
Test	FaceDancer	150	.958	144/150	.828	.601	107/150	.575
Test	TextDiffuserFT	149	.429	32/149	.456	.999	149/149	.718

ResNet’s 6/786 Digital-3 decisions and 32/149 TextDiffuser decisions reveal a much narrower test population for later evidence analysis than the pooled number suggests. TruFor detects every Digital-3 and TextDiffuser image here, but at its fixed threshold it also flags 169 of the 300 genuine test cards.

High AUROC on those families describes score ranking however it does not undo the false-positive rate. The large Digital-3 family can also dominate a pooled score while a smaller family behaves differently.

ResNet uses a forged-score threshold of 0.5. TruFor uses one threshold, 0.532956, selected on development images for every family and capture source. These classification thresholds are distinct from the requirements for acceptable spatial evidence. Table 4.4 shows the trade-off against TruFor’s upstream 0.5 operating point. The selected threshold improved apparent development accuracy and genuine specificity but missed five more forgeries. It was fitted to equal-weight image accuracy on these development images, not to a bank’s cost of accusing a genuine customer.

Table 4.4: TruFor development decisions at two fixed thresholds. The selected threshold was fitted on these same 459 images; the table is a calibration diagnostic, not independent validation. AUROC is unchanged because the score order does not change.

Forged-score threshold	Forged DC	BF correct	Accuracy	BA	Weighted F1
Upstream .500	293/306	51/153	.749	.645	.714
Selected .532956	288/306	73/153	.786	.709	.769

The one TruFor threshold also behaves unevenly across development capture sources (Table 4.5). For Huawei captures, only five of 51 genuine cards are accepted; the corresponding number for iPhone15Pro is 41 of 51. The groups are small and related through card identities. No device-specific threshold was fitted after seeing this spread.

Table 4.5: TruFor development results by capture source at the single selected threshold. Source-specific BA uses only that source’s forged and bona-fide images; each group contains 102 forgeries and 51 genuine cards.

Capture source	Forged DC	BF correct	BF specificity	BA
Huawei	99/102	5/51	9.8%	.534
iPhone15Pro	96/102	41/51	80.4%	.873
Scan	93/102	27/51	52.9%	.721

4.5 Does ResNet respond to the edited regions?

For ResNet, the development score were near perfect but test scores did not match up to that. Further, the development forgeries contain altered face and text regions, whereas the unseen families in test dataset change the mix of manipulations. The uneven decisions above raised a practical question of whether ResNet-18 learnt cues from both kinds of region edits(face, text), or mainly from one? A larger face edit might be easier for a whole-document classifier to learn than a small text change, but that is a hypothesis, not an explanation supplied by the detection scores. Hence the regional test was designed to examine that possibility.

For each suitable development image, the annotated face or text was replaced with the corresponding region from the same card’s bona-fide capture. Geometry and JPEG processing were matched. Figure 4.3 reports the average of ResNet’s forged-class scores *over the same paired cards* before and after replacement. This shows score movement even when both scores lead to the same class decision; it is neither a localisation score nor a calibrated probability of fraud.

On Digital-2 from development dataset, the mean score falls from almost 1.000 to 0.137 when the face is restored, but remains 0.998 when only text is restored. Restoring both regions brings it near the genuine-card score of 0.049. This supports stronger dependence on face-region content under the tested intervention.

It does not prove that text never matters or reveal the exact texture used. Replacement can also change local boundaries. Although the procedure matches JPEG processing and geometry, a separate classifier test of post-restoration artifacts was also performed. Its results evidence that post-restoration artifacts were not enough to separate out post-restoration artifacts. Altered boundaries still remain a residual risk in Appendix C. A

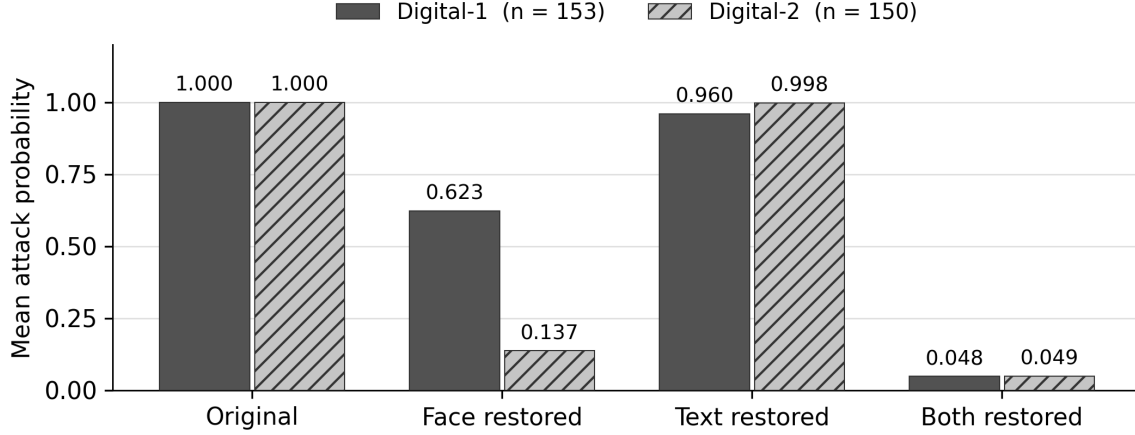


Figure 4.3: Mean ResNet forged-class scores after same-card regional restoration on development images. Digital-1 uses 153 pairs; Digital-2 uses 150 because three pairs have incompatible geometry. Bars describe paired image-score averages, not the effect of a single isolated forensic feature.

restoration-augmented training trial improved some internal cue balance but left official-test AUROC at 0.089 for Digital-3 and 0.463 for TextDiffuser. It was a diagnostic, not the primary model regardless.

Digital-3 from test dataset required another check. In 153 pairs, an *official-test Digital-3 forgery* was matched to the bona-fide parent card in this project’s development data. The forgery’s score rose relative to its own parent in 82.4% of pairs. Yet the average increase was only 0.0094 and its stem-resampled interval included zero; the same-source AUROC was 0.628 [0.592, 0.669]. The official-test AUROC of 0.059 instead compares all 786 Digital-3 forgeries with a different pool of 300 official-test bona-fide cards. These checks suggest that a weak response to the edit and a change in the comparison population could both matter. They do not identify a single proven cause, and the paired check was pursued after the official-test result was seen purely as diagnostics.

4.6 Agreement between maps and altered regions

Chapter 2 introduced altered-area fraction A (Area) and E as the Relevance Mass Accuracy (RMA). E is the fraction of map mass inside the annotated region; A is the fraction of valid document area it occupies. The ratio $\mu_w = E/A$ expresses how concentrated the map is relative to that area’s size. If a map spreads its mass evenly, E is close to A .

The Pointing Game (PG) asks only whether the *single strongest* map pixel falls inside an annotated region. Relevance rank accuracy (RRA) checks how many of the strongest pixels, selected in the same number as the altered pixels, lie inside it. Appendix B specifies the per-image calculations and invalid-map rules.

Table 4.6 keeps three denominators visible: the original forged family, its clean DC subset and the maps evaluated. All displayed map summaries are now restricted to DC. In particular, the archived ResNet FaceDancer audit covered all 150 maps; this table filters its individual-image records to the 144 correctly detected images, so its means differ slightly from the earlier whole-family description. ResNet excludes its artificial padding when calculating map agreement. TruFor uses its native manipulation map. A common-coordinate RRA audit still needs to be completed before setting the final evidence gate.

Table 4.6: Clean spatial agreement, conditional on a correct forgery decision. “All” is the original family size, “DC” the correctly detected count and “maps” the number summarised. A is altered document area in percent; E and PG are fractions. The displayed μ_w is the mean of per-image E_i/A_i , not the ratio of the displayed means. ResNet’s FaceDancer row is calculated from its per-image records after filtering to DC.

Map	Family	All n	DC n	Maps n	A (%)	E	μ_w	PG
Grad-CAM	Digital-1 (dev)	153	153	153	15.9	.707	4.77	1.000
Grad-CAM	Digital-2 (dev)	153	153	153	9.3	.546	6.16	.523
Grad-CAM	Digital-3 (test)	786	6	6	2.6	.0007	.03	.000
Grad-CAM	FaceDancer (test)	150	144	144	6.6	.494	7.82	.722
Grad-CAM	TextDiffuserFT (test)	149	32	32	4.3	.010	.24	.000
TruFor	Digital-1 (dev)	153	153	153	15.7	.555	3.90	1.000
TruFor	Digital-2 (dev)	153	135	135	9.3	.192	2.09	.504
TruFor	Digital-3 (test)	786	786	786	2.9	.642	23.26	.991
TruFor	FaceDancer (test)	150	107	107	6.9	.112	1.70	.402
TruFor	TextDiffuserFT (test)	149	149	149	4.2	.703	17.13	1.000

The “means” help describe a family, but can hide how varied its individual maps are. Table 4.7 gives the median and middle half of the image-level E values behind the preceding table. These are descriptive distributions for DC images, not DCEC pass rates.

Table 4.7: Median relevance mass E and interquartile range (IQR) among the DC images in Table 4.6. Values are computed from per-image map records; RRA thresholds are not inferred from this table.

Map	Family	Median E	E IQR
Grad-CAM	Digital-1 (dev)	.708	[.639, .784]
Grad-CAM	Digital-2 (dev)	.520	[.467, .624]
Grad-CAM	Digital-3 (test)	.00032	[.00005, .00150]
Grad-CAM	FaceDancer (test)	.531	[.367, .581]
Grad-CAM	TextDiffuserFT (test)	.00266	[.00031, .01199]
TruFor	Digital-1 (dev)	.550	[.406, .693]
TruFor	Digital-2 (dev)	.157	[.112, .232]
TruFor	Digital-3 (test)	.697	[.453, .859]
TruFor	FaceDancer (test)	.102	[.084, .124]
TruFor	TextDiffuserFT (test)	.778	[.579, .860]

An ID can have both a changed face and changed text. A high E over the *combined* annotation could still come entirely from the face. A PG success only establishes that one peak falls somewhere inside that combined area. Table 4.8 therefore separates the face and text contributions on the two development families. For Digital-2, ResNet places an average 0.540 of its total map mass in the face region but only 0.006 in text, even though its union E is 0.546. Its strongest point is in the combined annotation in only 80 of 153 cases. The component results expose what the union summary misses.

Table 4.8: Mean share of total map mass in the annotated face and text regions on correctly detected development forgeries. The union is the existing E measure; component results are shown because a union-level score need not cover both edits. Face and text annotations can overlap, so their shares need not add exactly to the union.

Map	Dev family	DC maps n	Face E	Text E	Union E
Grad-CAM	Digital-1	153	.658	.049	.707
Grad-CAM	Digital-2	153	.540	.006	.546
TruFor	Digital-1	153	.359	.197	.555
TruFor	Digital-2	135	.161	.031	.192

ResNet’s FaceDancer maps (test) show meaningful concentration on the edited area when restricted to the 144 DC images; its six correctly detected Digital-3 maps (test) and 32 TextDiffuser maps (test) show very little. Six Digital-3 examples cannot represent that 786-image family.

TruFor gives stronger clean spatial agreement on Digital-3 and TextDiffuser but weaker agreement on FaceDancer.

These differences describe the *implemented pipelines*, with different outputs and resolutions. The attack question compares each one to its own starting maps, so no claim that one architecture is inherently better is needed.

For the *mean E* values in Table 4.6, ResNet’s Digital-1 and Digital-2 development rows have document-stem-resampled 95% intervals [0.677, 0.736] and [0.513, 0.578], respectively. They describe sampling variation across these document stems for this fitted model. They do not include a new training seed or uncertainty in the annotation rectangles. Neither means nor medians can tell us how many individual maps pass both eventual *E* and RRA requirements.

4.7 Visual review and the unfinished evidence gate

The supplied rectangles approximate a manipulation, and an enlarged Grad-CAM map is coarse. A visual check is therefore needed alongside numeric agreement. The author is reviewing development maps against the altered document and its same-card bona-fide reference. Face and text are rated separately, with an ambiguous category where a yes/no judgement would overstate what can be seen. The ongoing coding task uses the displayed images and maps rather than per-image *E*, RRA, proposed DCEC membership or attack outcomes. Appendix B gives the coding sheet, sampling procedure and intended threshold-selection rule.

This is an *author-conducted contextual review*. The author has seen some aggregate results before rating the images, so it cannot be presented as an independent or prospectively blind assessment. Consistent display rules and a repeat rating of selected images can help expose inconsistent judgements; they cannot turn those judgements into ground truth or show how a document examiner would use a map. Ratings, agreement statistics and the final joint *E*/RRA thresholds remain incomplete at this version.

The order of work must also be disclosed. Attacks were first run on the broader DC population, before the clean-evidence rule was finalised. The remaining control is to choose the evidence rule from clean development measurements and the declared visual ratings, without optimising it against attack success. Official-test results, attack outcomes and a convenient eligible sample size must not select the thresholds. Prior analyst exposure remains a limitation. Appendix B records the sensitivity checks needed before a DCEC claim, including coordinate conventions, uncertain rectangle boundaries and component coverage.

4.8 What the clean population supports

Across the five manipulated families, ResNet has 488 observed DC images and TruFor has 1,330. These totals combine the development and official-test family counts in Table 4.3; they are *not* evidence-eligible sample sizes. For each pipeline, the eventual DCEC gate will retain DC images whose clean *E* and RRA both meet its accepted thresholds. The same accepted thresholds must then be used on that pipeline’s attacked maps. Until calibration

and coordinate checks finish, no DCEC coverage or shared-DCEC intersection can be reported.

The full denominator trail is original forged family $\rightarrow$ clean DC $\rightarrow$ clean DCEC. Low coverage at any stage will be reported, not hidden by reporting success only among eligible examples. Within the fixed DCEC set, DCEW means the attack makes spatial evidence fail while the forgery decision stays correct. A classification flip remains in the original DCEC denominator but is reported separately. The existing attacks on DC images can be analysed again under the later clean-only eligibility rule without rerunning them. Images eligible for both pipelines may provide matched visual examples and another view of *within-pipeline* degradation; their intersection is not a controlled architecture comparison.

4.9 Conclusion

The clean audit shows why a correct document decision is only a first step. ResNet’s very strong development detection does not carry evenly to unseen forgery families, and its Digital-2 map is concentrated mainly on the face. TruFor has different strengths across families and many false alarms on genuine official-test cards at its selected threshold. The compression and regional checks narrow some alternative explanations while leaving documented risks. Objective O1 now has a clear detection and continuous spatial baseline; O2’s author review and joint evidence calibration are still under way. Chapter 5 will assess adversarial evidence degradation against each pipeline’s fixed clean DCEC population once that population is established.

Chapter 5

Adversarial Evaluation of Spatial Evidence

5.1 What is being tested

Chapter 4 established that a correct forgery decision does not always come with a useful map. This chapter examines a different failure: whether a small, deliberate change to an image can make its map agree less with the annotated alteration while the detector still calls the image forged. The perturbation is applied *after* the forgery and after the controlled JPEG preparation. It does not repair or move the alteration.

The analysis begins with DC, the forged images that each pipeline called correctly before the extra perturbation. On these images, the paired change in relevance mass E measures how much of the map falls inside the annotated region before and after the attack. The stricter outcome in the research question starts with DCEC: images that also have acceptable *clean* spatial evidence under the joint E /RRA rule. An image becomes DCEW only when it was in that fixed DCEC set, its forged decision remains correct, and its adversarial E or RRA falls below the original evidence threshold. A classification flip is a separate outcome and stays in the DCEC denominator. Appendix E gives the recording rules.

The full ResNet-18/Grad-CAM attacks have been run on 450 DC images. They give a substantial paired result for E and a useful classification-attack control. TruFor’s attack settings have been frozen and its full run is in progress; an exact native-resolution gradient route and a one-image attack check have been completed. The clean human review, joint E /RRA thresholds and paired RRA results still need to be completed for both pipelines. For that reason, this chapter reports the measured DC results and leaves visible markers for the final DCEC-to-DCEW rates. It does not turn a fall in mean E into an unmeasured binary success rate.

The attacks are white-box: they use the model’s gradients and the altered-region annotations to seek a low E . This tests what is possible under a deliberately informed attacker. It is stronger access than a typical document submitter would have; the results concern these evaluated pipelines, not an observed attack on a live verification service. Work on explanation attacks shows why holding the label fixed matters, while document-localisation attacks provide a related, but different, forensic precedent (Ghorbani et al., 2019; Dombrowski et al., 2019; ?). Here loss of *annotation agreement*, rather than mere visual difference between maps, is the measured outcome.

5.2 Attack population and controls

Table 5.1 starts from the clean decisions reported in Chapter 4. Each family is counted separately. ResNet’s primary attack families were chosen after the clean audit: Digital-1, Digital-2 and FaceDancer had enough correct decisions and interpretable clean map agreement to examine degradation. Digital-3 had only 6/786 correct ResNet forgery decisions and near-zero clean E ; TextDiffuserFT had 32/149 correct decisions and mean $E = 0.010$ on those DC images. Calling further loss of their already poor evidence an adversarial explanation failure would obscure the more basic clean failure. This selection limits coverage and must remain visible alongside the attack result.

TruFor’s DC population is larger, but DC does not tell us how many maps will pass the eventual joint evidence rule. Its weak clean Digital-2 and FaceDancer map agreement is a particular reason to retain DCEC coverage by family. The two models operate at their previously selected decision thresholds: attack probability 0.5 for ResNet and forged score 0.532956 for TruFor. These thresholds remain fixed after seeing attack outcomes. Their image sizes, training exposure and map types also differ, so the main comparison is an attacked image against its *own pipeline’s* clean map, not an unqualified contest between models.

Table 5.1: Forged images entering the adversarial analysis. DC counts are clean, correct decisions; they do not establish DCEC eligibility. ResNet’s primary decision-preserving attack covers its three indicated families.

Split	Forgery family	Total	ResNet DC	TruFor DC
Dev	Digital-1	153	153	153
Dev	Digital-2	153	153	135
Test	Digital-3	786	6	786
Test	FaceDancer	150	144	107
Test	TextDiffuserFT	149	32	149
DC roster for evidence attack			450 [†]	1,330 [‡]

[†]Digital-1, Digital-2 and FaceDancer, completed. [‡]Frozen TruFor run roster across all five families; full population outcomes pending.

Across these development and test splits there are 1,391 forged images. ResNet called 488 of them correctly, but its primary evidence attack covers 450, or 32.4% of all 1,391. This is partly a consequence of poor clean detection on Digital-3 and TextDiffuserFT, not an adversarial exclusion chosen after seeing their attacked maps. TruFor’s 1,330 clean-correct cases cover 95.6% of all forgeries, although many may still have weak clean spatial evidence. These figures are useful coverage checks, not a like-for-like robustness score. Two ResNet attacked families are development families used in model selection; FaceDancer supplies the selected pipeline’s principal unseen-family attack result.

Both evidence-directed attacks minimise E within the union of the annotated altered regions while requiring the forged decision to remain correct. They use the same bound on changes to the RGB image, $\|\delta\|_\infty \leq 1/255$ in $[0, 1]$ units. Ten proposed gradient steps use a step size of $0.25/255$ and an eight-step feasibility pull-back when a proposal would cross the decision boundary. The attacks begin at the clean input without restarts. The unchanged Policy-C image is the baseline; no new crop or JPEG stage is applied during attack evaluation. For ResNet, only resized document-content pixels can change: its artificial side padding stays fixed. TruFor uses its full native-resolution document without padding. The precise model-space conversions and acceptance rules are in Appendix E; an L_∞ bound alone is not a perceptual study.

The ResNet classification-targeted PGD control asks a separate question: can a small perturbation change the *decision* to bona fide? Its result is necessary for interpreting Grad-CAM. When the attack-class decision disappears, a heatmap calculated for that class may also disappear; that cannot be counted as a decision-preserving spatial-evidence failure. This distinction follows the evaluation caution in §?: the attack and success condition must match the property being claimed.

5.3 ResNet: the classification-attack control

The classification control evaluated 456 images at the $1/255$ bound: 153 Digital-1, 153 Digital-2 and all 150 FaceDancer images. It pushed the score toward bona fide with ten projected steps and a random start. Among the 450 initially correct decisions, all 153 Digital-1, 153 Digital-2 and 144 FaceDancer decisions flipped. The six other FaceDancer images were already misclassified and are excluded from the conditional flip rate and the paired map-loss figures below.

Grad-CAM’s E also fell among these clean-correct images: the mean per-image relative losses were 66.75%, 97.72% and 86.70%, respectively. Yet the manipulated-class CAM had effectively zero positive mass on 30.7% of Digital-1, 96.7% of Digital-2 and 86.1% of FaceDancer adversarial images. The evaluator records $E = 0$ for such maps. It would be misleading to interpret those low E values as evidence being directed toward the wrong

part of a forged document: the forged classification was gone, and in many cases its positive map was gone too.

Three 72-image PGD pilots at 4/255, 2/255 and 1/255 each flipped all 70 images that were initially correct; two FaceDancer cases were already wrong. The full control then used the smallest of these *tested* conventional intensity-level budgets. No finer budget was tested, so the data do not locate a minimum perturbation at which decisions start to fail. They do show why the main evidence attack must enforce the original decision and check that its map remains defined.

This control has a practical implication for how the results should be read. At the tested bound, a white-box attacker targeting these selected ResNet images could simply seek a wrong forgery decision. The evidence-directed attack asks what can happen when the decision is *held* correct and a person might still inspect its map. Neither result says how often an uninformed submitter could achieve that condition or whether a downstream system would accept the altered file.

5.4 ResNet: reducing evidence while preserving the decision

The evidence-directed ResNet attack differentiates through Grad-CAM and seeks to move map mass away from the annotated union. It maintains the forged-minus-bona-fide logit margin at or above zero and only accepts a feasible proposal if E does not increase beyond a small numerical tolerance. The same frozen evaluator measured clean and adversarial maps after attack. All 450 final outputs stayed on the forged side of the decision threshold, all had nonzero positive CAM mass, and all 450 had strictly lower final E than their corresponding clean images. That last observation is an evaluated end result; the optimisation rule does not demand strict improvement at *every* accepted step.

Table 5.2 reports the most recent full visual rerun, consistently for all three families. The table pairs exactly the DC images that entered the attack. Its FaceDancer clean $E = 0.4940$ therefore differs from Chapter 4’s all-150 mean of 0.4888. The column headed “relative loss” is the mean of each image’s $100(E_{\text{clean}} - E_{\text{adv}})/E_{\text{clean}}$, not the percentage formed by dividing the two displayed family means.

Table 5.2: Completed ResNet decision-preserving attack on paired DC images. E and PG are clean→adversarial; relative loss averages image-wise losses. All attacked decisions remained forged and all adversarial CAMs were nonzero.

Family	DC n	Mean E_{clean}	Mean E_{adv}	Relative loss	PG, clean→adv
Digital-1	153	.7066	.1458	79.24%	1.000→.673
Digital-2	153	.5460	.1004	81.22%	.523→.235
FaceDancer	144	.4940	.0226	95.12%	.722→.000

Stem-resampled 95% intervals for the mean per-image relative loss are [77.86%, 80.59%] for Digital-1, [78.75%, 83.67%] for Digital-2 and [92.97%, 96.84%] for FaceDancer. These intervals respect repeated images from the same document stem. They describe variation from sampling stems within these selected DC populations; they do not include retraining the model, changing the rectangles or uncertainty about the later DCEC gate.

For Digital-1, mean E fell from .7066 to .1458 while the correct decision was retained on all 153 images. Its Pointing Game (PG) score, which checks only the location of the strongest map point, fell from 1.000 to .673. This difference matters: a maximum can still land in the altered area even when most relevance is elsewhere. On Digital-2, the mean was .5460 before and .1004 after attack. Its PG score fell from .523 to .235, though the clean result was already face-dominated: Chapter 4 found little map mass on the altered text. Lower union-level E cannot by itself show which field a reviewer would now overlook.

FaceDancer gives the strongest observed relative decrease. On its 144 clean-correct documents, mean E moved from .4940 to .0226 and PG from .722 to zero, with all 144 forged decisions preserved. Thus the classifier still gave the warning on these images while its Grad-CAM no longer put its single highest point in the annotated alteration on any of them. The result is stronger than saying the maps merely look different. It is also narrower than saying all 150 FaceDancer forgeries became misleading: six did not qualify as DC, and some of the remaining 144 may not pass the eventual clean DCEC rule.

The mean adversarial forged probabilities remained .920, .841 and .909 for Digital-1, Digital-2 and FaceDancer. These are reassurance that the final images were generally not balanced exactly on the decision line, but the individual decision check, rather than a family mean, establishes preservation. The visual rerun produced paired clean and attacked overlays for all 450 DC images. Producing panels is an audit aid; it is not a blinded judgment that a human examiner would make a worse decision. A figure-selection rule is given in Appendix E so later examples can show typical changes and counterexamples rather than only the most striking heatmaps.

The plot gives a second caution about averages. Individual FaceDancer losses were strongly concentrated near the top of the scale, while at least one was small; a large mean does not mean the search succeeded equally for every image. The three populations also have different clean map strengths and altered-area sizes. Dividing by each image’s own clean E makes the relative change readable, while absolute paired E and the original DC counts keep it grounded. Reporting only a percentage loss would hide whether the starting map had much annotated-region mass to lose.

These findings support one concrete claim: at this fixed budget and under annotation-aware white-box access, the evaluated ResNet classifier can remain correct while its Grad-CAM agreement with altered regions becomes much weaker. They do not establish a worst-case degradation over restarts, imply that benign image changes have the same effect, or prove that the classifier itself has stopped using the true alteration. Grad-CAM

is an attribution to the output; E measures agreement with the annotated region. Neither observation alone establishes explanation faithfulness or what a reviewer would infer from a particular map.

5.5 TruFor: full-resolution attack readiness and outstanding result

TruFor takes a native-resolution document and produces a manipulation-probability map and an image-level score. Its attack also minimises E under a hard correct-decision constraint; it does not optimise RRA directly. A routine full-image backpropagation ran out of GPU memory on the largest pilot documents. The adopted route recomputes selected network activations and evaluates attention in exact query chunks. It keeps the *whole image* in the calculation rather than resizing it or attacking tiles independently. The attack obtains gradients through that route, while a separate, unmodified TruFor instance supplies the reported maps and decisions.

This implementation was checked against ordinary computation where both routes fit in memory: the recorded worst map discrepancy was 9.54×10^{-7} and the gradient-sign agreement was at least .999998523. The largest tested image, about 6.78 megapixels, then completed a native-resolution backward pass. These are numerical and feasibility checks, not evidence that the full attack has succeeded on the study population. They reduce the risk of claiming resistance merely because the implementation could not obtain a gradient.

On one clean-correct TextDiffuser pilot image, one proposed step made a physical RGB change of only 0.25/255. Its annotated-region E changed from .755468 to .052624; the forged score went from .976692 to .850684, above the frozen .532956 threshold. This is a useful check that the objective, gradient and decision constraint can operate together. It is one image, selected for a memory stress test, with no verified RRA or DCEC status. It must not be turned into a population success percentage, an estimated DCEW transition, or a comparison of the two pipelines.

The ten-step TruFor protocol is now fixed for all 1,330 clean-correct forged images across the five families in Table 5.1. Its RGB budget and step size match the stated ResNet physical scale, with TruFor-specific conversion into its model coordinates. Execution progress has been recorded, but the accessible progress records contain work scheduling rather than the completed pairs of attack scores and spatial metrics needed here. A completed multi-step attack result and a TruFor classification-targeted control are therefore not reported at this revision. Attack outcome rates must be calculated from the final evaluated images, not extrapolated from the subset processed so far.

Chapter 4 shows why this gap matters by family. TruFor started with strong average annotated-region mass on clean-correct Digital-3 and TextDiffuser, but weak average

mass on Digital-2 and FaceDancer. If a family contains few clean DCEC maps, even a strong population-level DC attack could yield a small or unstable DCEC denominator. Conversely, the large Digital-3 clean-correct roster does not establish that its maps survive a bounded perturbation. The full paired distributions and the clean gate are needed to separate these questions.

[TRUFOR RESULT PENDING: insert the validated, full-run family-level paired clean/adversarial E , RRA and PG results, decision-preservation counts, nonzero-map checks, and stem-resampled uncertainty. State exclusions and failures, and include representative paired maps selected by Appendix E. Do not use execution-progress counts as an outcome denominator.]

The attack representation is continuous full-resolution RGB input to the model. Neither attack arm establishes that its effect would survive saving an 8-bit image or passing the adversarial image through another JPEG stage. That transfer question matters in an image-submission workflow and remains an explicit limitation of the present digital stress test.

5.6 From paired change to a DCEC-to-DCEW result

The measured ResNet falls in E show vulnerability among DC images, but E alone does not answer the complete research question. Clean DCEC membership requires $E \geq \tau_E$ and $RRA \geq \tau_{RRA}$ for each individual map. After that membership is fixed, DCEW requires the adversarial forged decision to remain correct and at least one of the two spatial tests to fail. The E objective may also change RRA, but no RRA transition should be claimed without measuring it. Conversely, an image with very poor clean evidence cannot count as a new DCEW failure even if the attack reduces its E further.

The clean-development visual coding and threshold calibration specified in Appendix B have not yet been completed, and TruFor’s paired RRA/common-coordinate checks are outstanding. Some attack work preceded the final design of that clean evidence gate. The remaining calibration must use clean development judgments without choosing thresholds to maximise an already observed adversarial effect; the chronology and the remaining risk should be recorded. Once the values are fixed, Chapter 4 will report the number of forged images progressing from total to DC to DCEC, by pipeline, split and family. This chapter will then report the count remaining decision-correct with acceptable evidence, the count changing from DCEC to DCEW, and the count flipping classification, all against the *same clean DCEC denominator*. Appendix E supplies the arithmetic and the table to fill.

[JOINT-GATE RESULT PENDING: insert frozen τ_E/τ_{RRA} and clean DCEC denominators from Chapter 4; calculate paired adversarial RRA, DCEC-to-DCEW counts and rates, separate decision flips, invalid maps and stem-cluster

intervals for each pipeline and family. Report overall coverage against the original family size. No numerical transition claim is made in this version.]

The main view remains within each pipeline: how far does it move from its own acceptable clean evidence? A second view can restrict attention to images that satisfy *both* pipelines’ clean DCEC rules, then examine a separate within-pipeline change on those same documents. Because one uses a classifier attribution and the other a native localiser, this common-image view controls the document sample, not the architectural or training differences. It should include actual image examples and its own denominator. The shared DCEC intersection has not been measured yet.

If the clean review does not support a stable threshold pair, the appropriate result is the continuous paired E /RRA change with the uncertain evidence classifications made explicit. A sharp DCEW percentage would imply more precision than the rectangles, map resolution and visual ratings support. The preselected E attack objective and the jointly evaluated E /RRA outcome must also stay distinct: an attacked map can lose mass without crossing the calibrated boundary, or cross it mainly through an RRA change. Those possibilities should be tabulated, not treated as contradictory measurements.

[SHARED-IMAGE VIEW PENDING: insert the intersection count by family, paired within-model changes, selection rule for visual examples and coverage relative to each pipeline’s full clean population after both evidence gates are fixed.]

5.7 Interpretation and remaining limits

The ResNet result establishes a large change in annotated-region Grad-CAM mass despite preservation of every selected correct forgery decision. Its classification control establishes a separate, serious susceptibility to label flips at the same tested budget, and shows why disappearing attack-class CAMs must be interpreted carefully. Taken together, these observations show the value of asking about the decision and its map separately. They are consistent with earlier work showing that stable predictions need not have stable explanations, while testing agreement with identity-document annotations under an explicit decision constraint (Ghorbani et al., 2019; Arras et al., 2022).

Several limits prevent a broader claim. The ResNet attack selects annotated regions, uses one clean starting point and a fixed ten-step search; it is a demonstrated vulnerability under those conditions, not a maximum over attacks. The development images and later diagnostics informed some design choices, so this is not a fully prospective test. Rectangular annotations and a coarse Grad-CAM can make a useful cue inside a face look poor by a coverage measure, or make a face-focused map look good while text is missed. The measured stem intervals address related captures for the DC-wide relative E result,

but not uncertainty in the uncalibrated evidence gate. Appendix E records these risks, what was checked and what remains.

This chapter therefore completes the DC-wide ResNet evidence and classification-control analysis, and establishes the TruFor attack’s technical feasibility and fixed design. The conditional DCEC-to-DCEW answer, its uncertainty and the shared-image view await the stated gate and attack results. Chapter 6 can synthesise the final two-pipeline result once those values are added; the present measurements already show why a correct forgery warning alone is insufficient evidence that its displayed map remains well aligned with the annotated edit.

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Appendix A

Method protocol, reproducibility and study-wide risks

This appendix makes Chapter 3’s analytical choices reproducible without filling the main chapter with file identifiers. Appendix B specifies the clean-map rating and threshold exercise; Appendices C and D hold the pipeline-specific clean decisions; Appendix E holds attack-specific checks and result templates. The rules here apply across those sections. “Pending” means that the procedure is specified but the evidence or calculation has not yet been verified.

A.1 Population and coordinate conventions

Use the public FantasyID open training release for the internal 160-stem project-train / 51-stem project-development separation. Link same-card captures by the dataset card stem *before* splitting. Record image filename, full stem, source device, bona-fide/forged status, manipulation family and split in the working manifest. The project development release of 459 images is an internal split of the 1,899-image open training release. Do not label it the dataset’s separately restricted validation release. Reserve the official test as a different image and manipulation-family source, while retaining the documented Digital-3 card-parent overlaps. The Digital-3 audit has 480 project-train parents, 153 project-development parents and 153 images without a matching parent; this does not establish the identity relations of FaceDancer or TextDiffuserFT.

Every manipulated image has one mask M formed by the union of its annotated altered rectangles after conversion to the relevant raster and clipping to valid pixels. If there are both face and text boxes, keep component masks as well as their union; do not double-count overlapping pixels. Define Ω as the actual document content on the working raster. For ResNet this is the resized content of the 512×864 canvas and excludes its zero-normalised horizontal padding. For TruFor it is the native image grid. Pixel counting

and top- $|M|$ selection take place inside Ω . For each tested map, save the transformed rectangles, $|M|$, $|\Omega|$, map-validity flag, E , RRA, PG and any tie diagnostic with the per-image record. Native rectangles need a review of 108 clips before the clean TruFor gate is claimed. If comparisons use a common display grid, keep the evaluation grid and interpolation rule stated separately from the illustrative resampling used to draw a figure.

A.2 Preparation and model construction

Policy C has two real JPEG operations with a decode between them: source-JPEG decode to RGB, encode at Q75/4:2:0, decode to RGB, encode at a deterministic matched final Q from 50–90/4:2:0, and decode the resulting file for a model. The assignment key is card stem plus capture source, so bona-fide and forged views of the same capture get the same final quality. Document the actual function used to assign Q and the cache manifest when making a reproduction package. JPEG-feature logistic-regression probes were fitted to project-training labels and measured on stem-distinct project-development labels before and after this control. Direct final-JPEG tables, broader history and the combined set are *distinct* feature sets. Report the class priors and the particular AUROC for each rather than claiming that all compression information disappeared.

The final ResNet training uses ImageNet-1K V1 initialisation, three frozen-backbone classifier-head epochs, then twelve end-to-end epochs. The recorded run uses seed 10, inverse-frequency class weights 1.5 (bona fide) and .75 (forged), AdamW weight decay .01, backbone learning rate 10^{-4} and head learning rate 10^{-3} . Select by project-development AUROC; the recorded selected full epoch is 12. These training settings specify the observed run, not a hyperparameter search with an independent validation set. The image height is 512 with original aspect ratio and central placement into 512×864 ; actual RGB uses the ImageNet mean and standard deviation, while the artificial side areas have zero value in the *normalised* tensor. The Grad-CAM target is the forged-minus-bona-fide logit at `layer4[-1]`; apply ReLU and divide by the positive map maximum if one exists, then interpolate to the full canvas. Measure spatial agreement within the document content only. Record the empty-positive-map case separately.

The final TruFor protocol retains its released weights. A cached Policy-C JPEG decodes to uint8 RGB, which is converted to model input by uint8/256. The manipulated-class probability of the native per-pixel softmax supplies the map; the separate sigmoid detector supplies the image score. A single detector cut-off 0.5329559743404388 was fitted for pooled image accuracy using 306 forged and 153 bona-fide project-development examples. The same cut-off is then used for every family and source, with no family-specific recalibration. Any resumed weights, alternate dev partition or older fine-tuned EP28 trial belongs to a separately labelled exploratory record.

A.3 Auditable analysis rules

A.3.1 Clean coverage and the joint gate

For every pipeline, split and forged family, log the original family size n_f ; the number called forged at the fixed clean decision threshold n_{DC} ; the number with valid clean masks and maps; and, when calibrated, n_{DCEC} . Retain bona-fide correct and incorrect counts separately for detection measures. Do not reduce the denominator of a detection rate because a forged map is invalid. Use the same two-threshold AND rule on all relevant images for a given pipeline. The *procedure* for fitting the two thresholds is shared across pipelines; the numerical values need not match.

1. Finish and lock the clean-development author coding table, preserving reviewer instructions, masked IDs, displayed panels, field-level 0/1/2 ratings and reasons for ambiguous images. Declare how those ratings become a binary visual reference, including the treatment of mixed face/text ratings, *before* joining metrics to labels.
2. Verify clean E /RRA on a consistent region and raster convention, examine top- K ties and clipped rectangles, and state the candidate grid and its tie-break rule. Select the pair that best fits the clean author reference by the declared balanced-accuracy criterion. This is within-development fitting, not external validation.
3. Inspect sensitivity to annotation dilation, component imbalance, stem-resampled recalibration and a few neighbouring candidate thresholds. Record the date, chosen thresholds, reviewed population and analyst exposure in a gate-freeze record. The ResNet attack was already measured at the DC level when this rule was designed; that chronology must be disclosed.
4. Apply the *frozen* thresholds to all relevant clean DC images and create the fixed DCEC ID lists. Only then join attacked per-image records and compute DCEW. If the clean calibration is too unstable to defend a binary gate, describe continuous paired changes and the instability instead of reporting a precise conditional transition percentage.

Examples of candidate E and RRA threshold grids are in Appendix B; none is a frozen cut-off here. An empty altered region, undefined rank, non-finite map or zero positive map cannot pass the clean joint gate. For attacked maps, state a primary convention for decision-preserved invalid maps *before* scoring and give a sensitivity table that keeps them separate. Do not set their continuous E to zero and merge them silently with well-defined, low- E maps. Record the number of tested eligible images that have no completed attack, since an unrun gradient step does not establish resistance.

A.3.2 Decision-preserving search: executable pseudocode

The following is pseudocode for the evaluated *method*, not a claim that all proposed TruFor outcomes have been validated. `AttackGradient` differentiates the attack map. `CandidateCheck` tests the fixed decision and objective E before accepting a step. For ResNet it uses the differentiable Grad-CAM objective, whose omitted positive max-normalisation cancels in E ; the final result is checked with the ordinary clean Grad-CAM routine. For TruFor the candidate and final checks use a separate unmodified model instance. C is the valid physical-RGB pixel support and T_m is that model’s fixed forged decision threshold. Project clips to RGB $[0, 1]$, to the clean-centred $1/255$ box and to the immutable ResNet padding when present.

```
input clean image z0, reference mask M, model m, valid support C
current = z0; (score, E, valid) = CandidateCheck(m, current, M)
require forged decision at threshold T_m; record clean map validity
if map is zero or invalid: mark the clean map outside DCEC;
log any DC-wide search under its checked implementation convention
for t = 1,...,10:
  g = AttackGradient(m, current, M) through chosen map
  proposal = Project(current - (0.25/255)*sign(g), z0, C)
  if CandidateCheck(m, proposal).decision is not forged:
    lo = current; hi = proposal
  repeat 8 times: mid = (lo + hi)/2;
  if CandidateCheck(m, mid).decision is forged: lo = mid
  else: hi = mid
  proposal = lo
  (newscore, newE) = CandidateCheck(m, proposal, M)
  if decision forged and newE is valid and newE <= E + 1e-8:
    current = proposal; E = newE
save current; ReferenceEvaluate verifies decision, E, map validity, bound
when paired RRA is implemented, evaluate it with the frozen gate
```

The sign and $0.25/255$ step refer to physical RGB. On ResNet, divide the per-channel RGB step and norm bound by its ImageNet standard deviation to perform the equivalent projection in the network tensor; restore padding unchanged. Because its CAM uses a first derivative, the E image gradient requires higher-order differentiation. On TruFor, the model tensor is $x = z\ 255/256$: its equivalent bounds are $1/256$ and $0.25/256$. Its full-image gradient uses activation recomputation and mathematically exact query-chunked attention rather than independent tiles. The separate TruFor reference model evaluates each candidate score and map. No extra score margin or score-drop tolerance is part of the feasibility rule in either arm. RRA, face/text components and PG are post-attack assessments, not terms in the attack loss.

ResNet’s classification-targeted control is a different projected-gradient search: target bona fide with cross-entropy, use one random start within the same physical $1/255$

box, then ten 0.25/255 steps. It is assessed against the initial clean decision, and cases already misclassified are kept visible. It cannot contribute to a decision-preserving DCEW numerator. A TruFor classification control and any additional perturbation budgets must be labelled as separate experiments when completed.

A.3.3 Counting, uncertainty and images

With $N_m = |\text{DCEC}_m|$ fixed from *clean* images, classify each eligible image in order: (1) no complete attack result; (2) final decision flipped, recording map validity as a separate flag; (3) decision preserved but map invalid; (4) decision preserved with valid map and failed evidence rule, n_{DCEW} ; or (5) decision preserved with valid map and passing rule, n_{stay} . These *exclusive* dispositions must sum to N_m . The main conditional attack fraction is n_{DCEW}/N_m , alongside N_m/n_f and absolute counts. An alternate tally counting decision-preserved invalid maps as evidence failures should reveal how much that choice matters; a flipped decision never counts as DCEW. For each same-document pair retain E_{clean} , E_{adv} , RRA on both when available, decision scores, validity flags and signed change; if $E_{\text{clean}} = 0$, relative change is undefined and must not be divided by zero.

For a 95% interval on a fixed family/pipeline statistic, resample card stems with replacement, retain all associated image pairs for a sampled stem and recompute the statistic on each replicate. Indicate the interval construction and number of replicates in the final reproduction record. A stem bootstrap respects repeated captures, but it neither resamples the classifier’s single seed nor corrects a biased family selection. For an interval on a DCEW proportion, keep the clean gate fixed and resample both numerator and eligible denominator; repeat the *clean-only threshold fitting* in a distinct sensitivity analysis if uncertainty in τ_E, τ_R is needed. Do not fabricate a confidence interval when N_m or attacked RRA does not yet exist.

Select illustrative overlays from the frozen sets by an announced rule: a case near median change, a low-change or unsuccessful case, a high-change decision-preserved case and a threshold-borderline case. Include a DC-but-not-DCEC map to show lost coverage, a two-component face/text example and at least one shared-image example if that intersection exists. Keep clean/attack maps on the same displayed colour scale where interpretation depends on intensity, and give numerical E/RRA , clean/attacked decision, family, split, mask and membership in the caption. These panels are examples, not evidence of the population frequency.

A.4 Method-wide risk register

Table A.1: Study-wide design risks. A completed check and a proposed check are distinguished; the right column states what the check cannot settle. More specific decisions remain in Appendices B, C and E.

Risk	Action and observed status	Remaining risk
Class-correlated JPEG history	Measured logistic probes before and after Policy C; matched final Q by stem/source; final-table AUROC .500.	Combined features still yield .586; re-encoding may hide authentic manipulation traces.
Related card captures inflate independence	Completed stem separation of internal train/dev; stem resampling for paired intervals; audited Digital-3 parent overlaps.	Official-test Digital-3 includes parents in internal sets; other-family identity relations are not asserted.
Poor clean decisions distort an attack headline	Report original forged count, clean DC, DCEC and attack outcomes by family; no DCEW label for a clean wrong decision.	ResNet’s Digital-3 and TextDiffuser clean coverage is low; conditional results have a narrow scope.
Broad mask or one large face field hides text misses	Measured area, mass and face/text components where available; planned joint rank gate and visual coding.	Rectangle boundaries, resolution and union dominance persist; TruFor RRA/common-coordinate audit is pending.
Gate fitted after some attacks	Mask per-image attack results from remaining clean review; select thresholds on development clean data; record decision chronology.	The researcher has already seen aggregate effects; selection is not fully preregistered or independent.
Spurious robustness from broken gradients	ResNet higher-order derivatives and frozen final checks; TruFor parity and full-image feasibility checks completed.	TruFor full paired results are unverified; one start/ten steps cannot establish a worst-case bound.
Decision flip mistaken for displaced map	Separate ResNet classification-targeted control and individual decision guard; check map validity.	No verified full TruFor classification control at this revision.
Benchmark effect treated as deployment effect	State GT-aware white-box access, processed-input norm bound, native/resized difference and continuous-float evaluation.	Effect after uint8 quantisation, fresh JPEG, camera recapture or on real IDs has not been established.
Selective examples or underestimated uncertainty	Predefine panel types; keep per-stem paired records and stem-cluster intervals.	One model seed, post-hoc diagnostics and annotation uncertainty lie outside a fixed-gate stem interval.
Dataset and ethics provenance not fully recorded	Fictional dataset and author-only coding described; approval/storage fields left for evidence check.	WMG approval/waiver, storage practice and training/confirmation documents must be verified before submission.

A.5 Reproducibility index and unfinished work

The study uses public Python-based ResNet/Grad-CAM and TruFor implementations, image and JPEG decoders, and the recorded model checkpoints. A reader needs the *actual* software versions, device record, file manifests and selected checkpoints, not just algorithm names. Table A.2 is a map to the current scripts in the two project repositories. Verify each path, run log and chosen artefact against the final dissertation submission before fixing an immutable version. Do not place digest values in the main thesis narrative; a complete digest/identifier manifest may be attached to this appendix or supplied as a linked reproducibility inventory.

Table A.2: Sources for a final reproducibility manifest. Paths are project-relative and need a checked commit/run link before submission. A path is evidence of implementation, not proof that a later experimental result has finished.

Decision or artefact	Script or result source to verify	Evidence currently available
Card split and ResNet fit	<code>scripts/train_resnet18_policy_c.py</code> ; selected-epoch training log.	Split assertions and selected full epoch recorded.
Policy-C preparation and compression probes	<code>scripts/prepare_policy_c_cache.py</code> ; <code>scripts/compression_leakage.py</code> ; <code>scripts/check_policy_c_leakage.py</code> .	Two-stage JPEG rule and probe scores recorded.
Clean ResNet Grad-CAM	<code>scripts/eval_resnet18_gradcam_clean_rma.py</code> .	DC-wide per-image clean audit; final joint pass rule pending.
Clean TruFor evaluator and score cut-off	<code>scripts/trufor/trufor_common.py</code> ; <code>scripts/trufor/03_calibrate_dev_threshold_accuracy.py</code> .	Native evaluator and global decision threshold recorded.
Shared-card Digital-3 audit	<code>scripts/audit_digital3_same_source.py</code> .	Parent-overlap counts; not a wholly unseen-card family.
ResNet decision-preserving visual run	<code>scripts/eval_resnet18_localisation_attack_preserve_cls_full_with_visuals.py</code> ; corresponding final visual log and per-image records.	450 validated pairs; avoid mixing with earlier run.
TruFor attack protocol and execution	<code>scripts/trufor/trufor_adversarial_attack_common.py</code> ; frozen attack configuration, Phase-A records and reference-evaluator logs.	Gradient parity/one-step feasibility checked; partial records ingested; full paired outcome pending.
Gate, ratings and final reporting	Dated rating sheet, mask-coordinate audit, gate-freeze record and final paired-score table: [AUTHOR VERIFY]	No frozen E/RRA pair or DCEW result yet.

[AUTHOR VERIFY REPRODUCIBILITY INVENTORY: repository/commit for each final code route; exact checkpoint and model weights; Python/library and JPEG-library versions; GPU/precision; stem split manifest; random seeds; Policy-C quality assignment; final ResNet visual run and complete TruFor run identifiers; paired-result row counts; any checksum/SHA256 values; and

Table A.3: Checks needed before the finished dissertation can claim the full conditional answer. “Pending” is not a measured zero.

Question	Present status / completion criterion
Are clean maps acceptable under a defensible rule?	Author face/text coding underway; validate TruFor RRA/native boxes and common coordinates, record the clean-development gate choice and robustness checks, then publish DCEC coverage.
Does TruFor’s ten-step attack change evidence at scale?	One-step full-image feasibility verified; evaluate and audit <i>all</i> selected per-image pairs or label each missing case and its cause.
Do clean-eligible maps cross into DCEW?	Join only a frozen DCEC roster to attacked E/RRA, scores and valid-map flags; give absolute counts, conditional rates and coverage for each family.
What does the same-image view show?	Compute the intersection of both clean DCEC rosters; report a separate within-model change for each pipeline on those images and show rule-selected panels.
Are approval and outputs reproducible?	Insert checked WMG approval/waiver and real data-management record; freeze final code/results/weights manifest with identifiers.

data-access licence or dataset version. Give sensitive or bulky identifiers in a linked manifest here rather than throughout Chapters 3–5.]

A.6 Redraw briefs for the three Chapter 3 figures

Chapter 3 contains typeset fallback figures, so the source compiles without separate artwork. For a final vector drawing, replace each fallback with the optional same-named PDF while retaining its caption and label. These briefs also state what a new drawing must avoid implying.

1. **Eligibility and outcomes.** Figure 3.1; optional file `Images/CH03_Eligibility_V3.pdf`. Draw a left-to-right flow from all forged images to the subset with a correct clean decision (DC), then to the subset meeting *both* clean *E* and RRA thresholds (DCEC). Branch the *fixed* DCEC denominator into incomplete run, changed decision, preserved decision with invalid map, preserved decision with failed evidence gate (DCEW), and preserved decision with passing gate. Show counts as symbolic *n* values, since the clean gate is not frozen. Note below the diagram that this is the order of *analysis*, whereas some attacks preceded gate design.
2. **Data and Policy C.** Figure 3.2; optional file `Images/CH03_Data_PolicyC_V3.pdf`. At the top show the 1,899-image open training release branching to project train (1,440 images, 160 stems) and internal development (459 images, 51 distinct stems). Keep the separate 1,385-image official test alongside; mark that 633 Digital-3 test

edits have a card/capture parent in project train or development. Below show source JPEG decode $\rightarrow$ Q75/4:2:0 encode $\rightarrow$ decode $\rightarrow$ stem/capture-matched final Q50–90/4:2:0 encode $\rightarrow$ decode. From the shared prepared image split into ResNet’s resized/padded transform and TruFor’s native-raster transform. Never label the official test identity-disjoint.

3. **Attack arms.** Figure 3.3; optional file `Images/CH03_Attack_Arms_V3.pdf`. Draw parallel branches from the prepared forged image. ResNet’s branch changes document-content pixels on a 512×864 canvas, differentiates through forged-class Grad-CAM and checks logit margin ≥ 0 ; TruFor’s branch changes native pixels, differentiates through its manipulation map and checks image score ≥ 0.532956 . Beneath both put the common physical RGB L_∞ budget $1/255$, ten $0.25/255$ steps, eight feasibility bisections if needed, and a final reference-evaluator check. Draw RRA as an *afterwards* measurement. Label the completed ResNet DC sample $n = 450$ and TruFor’s frozen DC roster $n = 1,330$ as populations, with TruFor full paired result pending. Do not portray either as DCEC or DCEW.

Appendix B

Spatial measures, human review and calibration

B.1 How an image enters the clean evidence audit

The unit for the evidence rule is one manipulated image and one pipeline. An image enters DC if that pipeline calls it forged on the clean input at its fixed *decision* threshold. Bona-fide images still inform specificity but have no annotated altered region, so they do not enter the manipulated-region E/RRA gate. Repeated captures of a document remain associated through its stem. The number entering each stage must be reported by split and manipulation family: evaluated forged images, DC, and eventually DCEC. A percentage conditioned on DCEC must always be accompanied by the size of the original family.

For a valid map $S_i(p) \geq 0$, altered-region union M_i and valid document support Ω_i , the measures are

$$\begin{aligned} A_i &= \frac{|M_i|}{|\Omega_i|}, \\ E_i &= \frac{\sum_{p \in M_i} S_i(p)}{\sum_{p \in \Omega_i} S_i(p)}, \\ \mu_{w,i} &= \frac{E_i}{A_i}, \\ \text{RRA}_i &= \frac{|\text{TopK}_{|M_i|}(S_i) \cap M_i|}{|M_i|}. \end{aligned} \tag{B.1}$$

PG is one when the strongest valid map pixel lies in M_i , zero otherwise. Intersections, ranks and maxima must all be evaluated within Ω_i . These definitions agree with the mass/rank distinction explained in Chapter 2 (Arras et al., 2022; Wang et al., 2020). The combined *E/RRA pass rule* is this study’s own operational criterion, not a standard threshold supplied by either paper. Report per-image distributions as well as means and bootstrap intervals; $\text{mean}(E_i/A_i)$ is not $\text{mean}(E_i)/\text{mean}(A_i)$.

The masks union all altered rectangles after clipping to valid image bounds. Do not include ResNet’s artificial padding in Ω or in the top- K ranking. TruFor’s existing outputs are measured on its native pixels, whereas the established ResNet audit measures the resized document inside its padded canvas. The final paired/common-image analysis needs a recorded common content-coordinate convention, including interpolation, clipped-box rounding and an audit that rankings and memberships remain coherent across resolutions. Merely naming the same metric does not make the two current rasters identical.

An empty region, zero total map mass, NaN, or other invalid map needs an explicit diagnostic category rather than being silently assigned a favourable score. For ties at rank K , record the lowest and highest possible RRA from equally ranked pixels. Use a fixed coordinate rule for the reported value. A preliminary ResNet development check recorded 0/306 membership changes at provisional RRA 0.5; its original output still needs to be linked to the final evidence record. This says nothing about later thresholds or TruFor ties.

B.2 Visual review to support the evidence rule

Status at this revision: the author has begun reviewing the prepared panels. A completed, locked coding table and measured human–metric agreement are not yet available. The following is the protocol to complete and document; its suggested codes and candidate thresholds must not be written up as achieved results. Because the author has seen some aggregate findings, this is a contextual review with per-image outcomes withheld from the coding task, not independent blind validation.

Create one record per clean DC development image and pipeline, with a stable but masked review ID. Default coverage is all 306 ResNet development DC forgeries and 288 TruFor development DC forgeries; if capacity requires sampling, select strata by family, annotated area and preliminary map behaviour *before* opening the ratings, then disclose the sampling fraction. Present each document at readable native resolution with the altered-image overlay and its corresponding clean bona-fide reference. Show the annotated rectangle or the difference only where needed to make the rating task clear, using the same display convention for both pipelines. Preserve a copy of the exact viewed panels and instructions.

Hide all E, RRA, μ_w , PG, candidate acceptance status, attack outcomes, filenames that reveal an experiment outcome and any proposed numerical threshold. Randomise order within pipeline, and avoid displaying two outputs for the same stem consecutively. A reviewer may still recognise the family or the model from the map’s appearance. If the researcher has already seen earlier attack summaries, the later ratings can be *blinded to the displayed values* but cannot honestly be described as prospectively uninformed. Record

who rated, when, whether any images were rerated and any earlier familiarity with the material.

For the two development families, which each alter face and text, use separate ordinal face/text grades: 0 (no meaningful localisation), 1 (partial or uncertain coverage), 2 (clear coverage). The short codes in Table B.1 preserve where a union-level pass might conceal neglect of one field. They are proposed labels for the coding sheet, not observations.

Table B.1: Proposed clean-development visual codes for images containing both face and text edits. The reviewer records a brief reason when using an ambiguous rating.

Code	Face	Text	Interpretation
BY	2	2	Both altered components clearly indicated.
FFPT	2	1	Face clear, text partially indicated.
PFFT	1	2	Face partial, text clear.
F	2	0	Face indicated, text missed.
T	0	2	Text indicated, face missed.
N	0	0	Neither component meaningfully indicated.
50	–	–	Genuinely ambiguous; reason retained.

For the primary *union-level* human reference, one proposed rule treats BY, FFPT and PFFT as acceptable; F and T are component-only and remain unacceptable for the main binary comparison, while 50 is set aside. Test the alternative in which F and T also count as acceptable, because the answer may depend on how a user interprets the union of altered regions. If 1/1, 1/0 or 0/1 combinations occur, record their literal grades and a reason rather than forcing them into an inaccurate short code. For one-component official-test manipulations, grade only the component present; those later images provide examples and external checks, not labels for fitting the development thresholds.

The rating task asks whether a map visually agrees with an annotation. It is not a study of reviewers’ decisions in a bank, an explanation-faithfulness test, or a pixel-exact manipulation boundary assessment. A repeated masked subset can estimate the same reviewer’s consistency; a second reviewer is useful if available, with disagreement and adjudication recorded rather than concealed. Do not report agreement statistics until ratings actually exist.

B.3 Joint threshold selection and what remains to be checked

Join reviewed development IDs to clean per-image E/RRA only *after* ratings are locked. For each candidate pair $(\tau_E, \tau_{\text{RRA}})$, call an image spatially acceptable if both inequalities hold. Compare this predicted status with the declared human reference. A possible search grid is $\tau_E \in \{.40, .45, \dots, .70\}$ and $\tau_{\text{RRA}} \in \{.50, .55, \dots, .80\}$, but the actual candidate grid

and tie rules must be recorded before fitting; this line is a proposal, not a chosen pair. Use the same selection procedure for both pipelines while allowing different numeric thresholds because the map distributions differ.

The proposed main fit criterion is balanced accuracy against the human labels, so an imbalance between visually acceptable and poor maps cannot make ordinary accuracy look artificially strong. Record sensitivity, specificity, confusion counts, precision and agreement by Digital-1/Digital-2 as diagnostics. Examine whether RRA excludes images that E alone would accept, and whether each threshold disproportionately removes a family or a component type. Never optimise thresholds against official-test results, adversarial E/RRA or a desired attack-success rate. Because some attacks existed before this protocol was finalised, maintain an explicit audit of data actually viewed during selection; a clean-only computation does not erase analyst exposure.

Perturb the rectangles outward by small fixed distances in a defined common coordinate system, and recompute area, E, RRA and eligibility. The proposed 8- and 16-pixel dilations are measured on a 512-pixel-high reference document and converted proportionally when the native raster differs. Report the fraction of borderline images changing membership, not just new pooled means. Resample document stems and refit the entire candidate grid to describe threshold instability. Review top- K ties again at the finally selected RRA threshold. Provide separate face/text E and, where feasible, RRA so the union criterion’s limitations remain visible. If these checks reveal no stable joint cut-off, report that limitation instead of forcing a binary DCEC headline.

Table B.2: Evidence available for the final DCEC gate. “Pending” means the appendix gives a procedure, not a completed result.

Required item	Current record
ResNet dev per-image E/RRA and provisional tie audit	Available; 0/306 sensitive at RRA 0.5.
TruFor E/area/PG on native maps	Available; matching RRA and common-coordinate check pending.
Author visual labels and reviewer consistency	Rating underway; completed codes and agreement pending.
Joint threshold values and human-agreement results	Not frozen or reported.
Rectangle dilation and component RRA	Dilation pending; face/text E exists for ResNet, component RRA pending.
Final DCEC counts and shared-image intersection	Pending on completed preceding checks.

Once the evidence pair is selected, retain the exact development population, annotation convention, reviewed labels, selection criterion and threshold values in a dated freeze record. The figures and results in Chapter 4 are descriptive DC baselines until that record exists. The measured DCEC size, continuous spatial distributions and joint cut-offs then

belong in Chapter 4; any DCEC-to-DCEW rates belong in Chapter 5. Model and code identities, run filenames, checkpoint hashes and content digests belong in the separate reproducibility Appendix D.

B.4 How visual examples will be selected

When the screenshots are ready, select examples by a recorded rule rather than choosing only attractive heatmaps. For each pipeline include a clear two-component development example, a case with high E but low RRA or an ambiguous map, a face-focused/text-poor case where applicable, a low-agreement DC image and an image near each final threshold. Add at least one official-test single-component example. Show a document crop at readable scale, the map with the rectangle overlay, the family and whether the image belongs to DC or DCEC, but conceal personal-looking details if an image is later shared beyond the dissertation. Report the values A, E, RRA, PG once calculated, making clear that these are illustrative cases, not a representative performance estimate.

Appendix C

ResNet-18 clean-audit decisions and risks

C.1 Preprocessing and chosen clean model

The analysed model is the compression-controlled, whole-document ResNet-18 trained on project-train Digital-1 and Digital-2 with disjoint development stems. Policy C re-encodes at JPEG Q75 and then a matched deterministic final Q50–90. The model resizes document content to a height of 512 while preserving aspect ratio, centres it on a 512×864 canvas and excludes the padded margins from spatial measures. Grad-CAM uses the final residual block and the forged-minus-bona-fide logit. Its map is coarse, approximately 16×27 before interpolation. Because this is a classifier attribution, a correct class decision and high Grad-CAM annotation agreement answer different questions.

The initial raw JPEG-only development AUROC was 1.000. Under the adopted control, final table features alone gave 0.500, history 0.524 and combined compression features 0.586 [0.572, 0.606]. These numbers justify treating the perfect controlled ResNet development AUROC as a result needing additional checks, not proof that it identifies every manipulation. The held-out development recall was 306/306 forged images; specificity was 148/153 bona fide images. The official-test diagnostic had just 6/786 Digital-3 and 32/149 TextDiffuser correct decisions, but 144/150 FaceDancer correct decisions. Chapter 4 keeps these family denominators visible.

C.2 Alternative explanations tested

Same-card, same-capture regional restorations provided a content intervention. For 153 Digital-1 and 150 compatible Digital-2 pairs, restoring the face reduced mean forged scores to 0.623 and 0.137, respectively; restoring both face and text reduced them to roughly 0.049. Text-only restoration left Digital-2 near 0.998. This supports face-dominant decision

evidence under the intervention, but matching and blending do not prove the specific feature learned. The excluded three Digital-2 geometry mismatches must not be silently counted in the 150 paired cases. Figure 4.3 shows the results in the main chapter.

The official Digital-3 AUROC of 0.059 against the official bona-fide pool prompted matched-source diagnostics. For 153 official-test Digital-3 edits paired with their project-development bona-fide parent captures, scores rose after editing in 82.4% and the matched AUROC was 0.628 [0.592, 0.669]. The mean rise was small, 0.0094, with a 95% interval including zero. A further test retained about 91% of gross greyscale pixel difference after resizing and Policy C. These findings weaken the single explanation that the edit reverses the detector’s signal or is simply erased by resizing, without proving that it learns a particular alternative cue. A native text-patch variant improved the same-source Digital-3 diagnostic to AUROC 0.740 but did not solve official generalisation. It was not substituted for the primary whole-document model.

C.3 ResNet-18 risk register

Table C.1: ResNet clean risks. “Measured” identifies an actual test; a proposed control is labelled as such. The residual risk is part of the finding.

Risk	Mitigation and current evidence	Remaining limitation
Class-correlated JPEG processing	Measured Policy-C compression-feature probes; direct final-table AUROC .500.	Residual combined-feature AUROC .586; recompression changes traces.
Correct decision from irrelevant cues	Measured same-parent restoration with matched JPEG processing and geometry; face dependence in both development families.	A separate classifier result for restoration artifacts has not been located; patch boundaries and other cues remain possible.
Misleading official-test reversal	Measured same-source Digital-3 scores and 153 matched pairs.	Source population and forgery family still shift together in the official comparison.
Coarse map hides text failure	Measured E and face/text E separately; Digital-2 mean text E only .006.	Component RRA, visual grading and GT dilation remain pending.
Overconfident clean eligibility	Report full-family and DC counts before eventual DCEC counts.	Evidence cut-offs and DCEC coverage have not been frozen.
Related captures understate uncertainty	Measured stem-resampled intervals for selected clean summaries.	One trained model/seed and uncertain rectangle boundaries are outside those intervals.

Area summaries help interpretation: Digital-1 mean annotation area is 15.9% and mean E is 70.7%, while Digital-2 area is 9.3% and E is 54.6%. The respective mean area-normalised concentrations are 4.77 and 6.16, computed image by image. On Digital-2, about 98.8% of the mass *inside the annotated union* belongs to the face region. Thus a union-based gate may accept face-focused evidence while still failing to help a reviewer locate a text edit. Face/text breakdowns and a blinded visual review should accompany any final union-based DCEC count.

Appendix D

TruFor clean-audit decisions and risks

D.1 The evaluated zero-shot condition

The current native pipeline uses pretrained TruFor weights without FantasyID fine-tuning, on Policy-C full-resolution images. Its manipulated-class localisation probability is the spatial map for E, A , μ_w and PG; the separate confidence information participates in the model’s pooled image decision but is not substituted for the localisation map. One global forged-score threshold, 0.532956, was selected on all 459 development images by pooled image accuracy and then used across families and devices. It is a completed *classification* calibration, not the pending evidence E/RRA calibration.

At that decision point, development forged recall is 288/306 and bona-fide specificity 73/153. Official-test forged recall is 1,042/1,085, while specificity is 131/300; 169 bona-fide images are false positives. Family detection and DC-spatial results appear in Tables 4.3 and 4.6. Native-map E is especially high for correctly called Digital-3 (0.642) and TextDiffuser (0.703), but low for clean-correct FaceDancer (0.112) and Digital-2 (0.192). These are means, not estimates of how many individual maps meet a future joint gate.

The development calibration split is also the reported development evaluation split, so its apparent 78.6% accuracy is not an independent check of calibration. The optimisation treats every image equally in a 306-forgery/153-bona-fide pool. It does not represent bank-specific error costs or cure the low official-test specificity. Huawei development bona-fide specificity was only 9.8% under the same global threshold. No post-hoc device-specific adjustment was adopted.

D.2 Historical adaptation and evidence status

An older fine-tuned TruFor EP28 experiment exists, so it would be inaccurate to say the project never attempted adaptation. That experiment used an earlier development split and a different evaluation protocol. Its report gives perfect development classification

measures but official-test AUROC 0.409 and 0/786 correct Digital-3 decisions. It is an adaptation diagnostic showing that encouraging development performance did not establish transfer; its numbers must not be merged with, or treated as a controlled ablation of, the later Policy-C zero-shot run. The main Chapter 4 tables describe only the latter.

Existing TruFor runs generated spatial panels, but the panel count is not a completed blinded rating study. The native-mask implementation clipped 108 altered rectangles to image bounds; these need a case-level review to distinguish expected edge clipping from annotation-coordinate errors. A common content-coordinate E/RRA evaluation with ResNet, the human-review labels and boundary sensitivity remain open. Until those checks are complete, no TruFor DCEC count or cross-pipeline DCEC intersection is established.

D.3 TruFor risk register

Table D.1: TruFor clean risks, distinguishing measured controls from work still needed for the evidence gate.

Risk	Mitigation and current evidence	Remaining limitation
Decision threshold confuses ranking with operating quality	Measured one global dev-accuracy threshold; report recall, specificity and AUROC separately.	Calibration and evaluation share development images; test false positives remain high.
Map and score have different roles	Measured the native manipulated-class probability as spatial output; detector score used for DC.	High document score does not imply acceptable map agreement.
Device/family shift	Measured family- and device-stratified clean results without a per-device refit.	Weak FaceDancer E and Huawei specificity constrain interpretation.
Annotation and coordinate error	Rectangles clipped to native bounds and retained in the current E audit.	Audit of 108 clipped rectangles and common-coordinate RRA still pending.
Prepared maps mistaken for human validation	Author rating is under way; Appendix A specifies masking of per-image outcomes.	No locked labels, agreement statistics or joint thresholds.
Fine-tuning trial mistaken for like-for-like comparator	Historical EP28 result labelled as a different split/protocol.	It cannot isolate the causal benefit or harm of adaptation in the current setup.

The primary future comparison is still within TruFor’s own clean DCEC set. A shared eligible-image subset can help illustrate individual examples once the two evidence gates exist, but different training exposures and spatial outputs remain part of that interpretation. Full run provenance, implementation versions and checkpoint records can be placed in the separately planned reproducibility Appendix D.

Appendix E

Adversarial evaluation: protocols, evidence and risks

E.1 Attack question and fixed counting rules

This appendix supports Chapter 5. A clean forged image i enters DC_m for pipeline m only when the previously fixed decision rule calls it forged. The clean evidence rule then fixes a subset, independently for each pipeline:

$$\mathcal{D}_m = \left\{ i \in \text{DC}_m : E_{i,m}^{\text{clean}} \geq \tau_{E,m} \wedge \text{RRA}_{i,m}^{\text{clean}} \geq \tau_{R,m} \right\}. \quad (\text{E.1})$$

The joint cut-offs are to be calibrated from *clean* development maps and masked human ratings under Appendix B, and then fixed. They are currently pending. The two pipelines follow the same selection procedure; a numerical cut-off may differ because the two kinds of map have different distributions. $\mathcal{D}_m$ must not be reselected after the adversarial run. Images outside $\mathcal{D}_m$ can still contribute to the descriptive DC-wide E analysis, but cannot undergo a DCEC-to-DCEW transition.

Among $\mathcal{D}_m$, the adversarial outcome is DCEW only when the original forged decision remains correct and $E_{i,m}^{\text{adv}} < \tau_{E,m}$ or $\text{RRA}_{i,m}^{\text{adv}} < \tau_{R,m}$. With $N_m = |\mathcal{D}_m|$, the conditional transition rate is

$$\hat{r}_{m,\text{DCEW}} = \frac{\sum_{i \in \mathcal{D}_m} \mathbf{1}\{\text{DC}_{i,m}^{\text{adv}} \wedge (E_{i,m}^{\text{adv}} < \tau_{E,m} \vee \text{RRA}_{i,m}^{\text{adv}} < \tau_{R,m})\}}{N_m}. \quad (\text{E.2})$$

Report separately the N_m images that remain DCEC after attack, the images that flip classification and the images with missing or invalid attacked maps. A flip never counts in the DCEW numerator, even if its map has little mass. For an undefined map, agree and record a handling rule before final scoring: an undefined E or RRA is never an acceptable evidence pass, but it must not be silently assigned a favourable value or mixed into the numerical E distribution. State whether the primary DCEW tally treats a decision-

preserved invalid map as evidence failure, and provide the alternate tally with invalid cases separated. This choice is outstanding until the gate and invalid-map convention are frozen.

For each family, display $n_{\text{forged}} \rightarrow n_{\text{DC}} \rightarrow N_m$ before an attack rate. A percentage among N_m is conditional coverage, not a percentage of all forged images. For each family and pooled results, report paired clean and adversarial E and RRA, absolute change, and the mean of per-image relative E changes where $E^{\text{clean}} > 0$. A ratio of two family means is a different statistic. Retain signed individual differences so images that improve are not dropped from a family summary. PG and component-level face/text maps help interpret the result but are not substituted for the joint gate.

E.2 ResNet-18: methods and checks

The selected classifier is the Policy-C ResNet-18 and final-block Grad-CAM described in Appendix C. For each annotated forged image, altered boxes are projected onto the 512×864 classifier canvas; Ω is the resized document-content part of that canvas, without side padding. The adversary changes RGB content pixels only. Resizing precedes the perturbation; the measured norm bound refers to these RGB canvas values, not to an already exported file or to original native-image pixels. All outputs are evaluated with the same map and region geometry as their clean baselines.

The classification control uses targeted projected gradient descent toward bona fide, a random start, ten steps, step size 0.25/255, and an RGB L_∞ bound 1/255. It evaluated 153 Digital-1, 153 Digital-2 and 150 FaceDancer images; at the final decision threshold of 0.5 it flipped 153/153, 153/153 and 144/144 of those initially DC. The six already incorrect FaceDancer cases are not part of the conditional map-loss figures. Three pilot budgets 4/255, 2/255 and 1/255 each flipped 70/70 initially correct images among the same 72 pilot inputs; 1/255 was the smallest tested conventional budget, not a certified minimum. The respective final zero-positive-CAM fractions *among DC images*, 30.7%, 96.7% and 86.1%, make E under this control hard to interpret as a displaced still-positive explanation. For near-zero content CAM energy the frozen control evaluator records $E = 0$, PG=0 and leaves map-shape comparisons undefined.

The evidence attack instead differentiates E through the Grad-CAM gradient weights; this requires derivatives through a derivative. The following is a compact account of the implemented search, not a separate proposal for future code:

1. Start from the clean image; compute the reference decision and E against the altered-region union.
2. Propose a sign-gradient step of size 0.25/255 toward smaller E ; clip RGB to its valid range and to the clean-centred 1/255 box; restore all padding exactly.

3. If the forged-minus-bona-fide logit margin would become negative, bisect from the last feasible image toward the proposal for eight steps to recover a feasible point.
4. Accept a feasible proposal when its evaluated E is at most the current $E + 10^{-8}$; otherwise retain the last feasible image. Repeat for ten proposals. Evaluate the final image again using the frozen clean evaluator.

The small tolerance means an accepted step is feasible and *approximately* non-worsening, not necessarily strictly improving. The observed final E was strictly below clean E for all 450 images. There was one clean start, no random restart and no search over step size. As an objective, E considers the altered union; it does not independently force coverage of both an altered face and altered text or optimise RRA.

The final visual rerun is the sole source for the Chapter 5 attack means. The earlier nonvisual full run had slightly different final E values; for example its Digital-1 attacked mean was about .1466 rather than the final .1458. The source of that numerical difference has not been resolved. Both support the same large decrease, but mixing one run’s clean/attacked pairs with the other’s relative change would be unjustified. Final reported mean forged scores were .920 for Digital-1, .841 for Digital-2 and .909 for FaceDancer; every evaluated decision was correct and every attacked positive CAM nonzero. Decision checks were performed for each individual, not inferred from these mean scores.

The most important interpretation check remains the clean baseline for each attacked population. FaceDancer’s paired 144-image DC clean E is .4940 and PG .722, whereas the descriptive all-150 clean row in Chapter 4 is $E = .4888$ and PG .700. Neither all-150 number should be used to calculate an attack effect on the paired 144. The two development families have 51 stems each; the 144 FaceDancer DC images cover 86 stems. The completed full visual run reports 95% document-stem bootstrap intervals for mean per-image relative E loss: Digital-1 79.24% [77.86,80.59], Digital-2 81.22% [78.75,83.67] and FaceDancer 95.12% [92.97,96.84]. Paired RRA and DCEC-to-DCEW intervals cannot be reported until those metrics and their clean denominator are established.

E.3 TruFor: full-image method and checks

The TruFor attack uses the Policy-C native-resolution image and the frozen zero-shot model from Appendix D. Its decision is forged when the image score is at least 0.5329559743404388. The objective is to lower E on the native manipulated-class probability map, measured against the altered-region union; it does not target the image score downwards and does not optimise RRA as an objective. The frozen ten-step search has the same stated physical RGB 1/255 limit, step 0.25/255, clean start, no restarts, eight boundary-feasibility bisection steps and acceptance only if the image remains forged and $E_{\text{candidate}} \leq E_{\text{current}} + 10^{-8}$.

Table E.1: ResNet attack risks. Checks already performed are distinguished from work needed to narrow the interpretation.

Risk	Measured mitigation or decision	Residual limitation
Classification failure masquerades as evidence failure	Separate targeted PGD control; decision-preserving attack checks each forged decision and positive CAM.	The same 1/255 budget also flips every selected DC case under the control.
Padding or norm violations explain the effect	Content-only projection; explicit RGB budget and zero padding-change checks.	Perturbation occurs on the classifier canvas, not a re-exported native image.
Uninformative clean maps enter the attack headline	Restrict primary attacked families by clean detection and maps; publish excluded-family counts.	DC-wide E falls are not DCEC-to-DCEW rates; coverage can become smaller after the gate.
GT-aware objective exaggerates realistic access	State knowledge of exact altered rectangles and use the same union for clean/attack metrics.	A document submitter may not know the examiner’s annotations or detector gradients.
Fixed search understates achievable damage	State single start, fixed step size and ten proposals.	No worst-case guarantee, nor guarantee that another search would fail/succeed.
Area/geometry misstate map usefulness	Keep PG and region-specific context; apply the planned clean gate/visual review.	Coarse Grad-CAM, broad boxes and weak text evidence remain; dilation and RRA checks pending.
Run drift/related images overstate precision	Use one final rerun for all Chapter 5 values; report paired stem intervals for relative E loss.	E07/E08 difference unexplained; RRA/DCEW intervals not yet available.

The clean input is held in continuous float32 for optimisation. If the physical RGB $[0, 1]$ input is z , the model uses $x = z \cdot 255/256$, so the physical 1/255 limit and 0.25/255 step correspond to 1/256 and 0.25/256 in model coordinates. This distinction avoids accidentally giving TruFor a different pixel budget. All native document pixels can change because this model has no artificial padding. The authoritative attacked object is the continuous model input; there is no uint8 requantisation or second Policy-C JPEG pass before attack evaluation.

The full-input gradient originally exceeded available memory. Activation recomputation in the noise branch and transformer MLPs, plus exact splitting along the attention query axis, enable the whole-document gradient without tiling or approximating attention context. Where a conventional graph fitted, the recorded worst map discrepancy was 9.54×10^{-7} , gradient cosine 1.000000000 and worst gradient-sign agreement .999998523. The 6.782-megapixel largest pilot input completed the recomputed backward route, with all 32 attention modules split. A separate unmodified TruFor model evaluates each attack candidate’s score and map; attack-engineering changes therefore do not define the reported spatial outcome. This checks a likely gradient-obstruction failure mode, but does not make a one-image result representative.

In that one-image native TextDiffuser check, E fell .755468→.052624 after one 0.25/255 physical step; the forged score remained above the threshold at .976692→.850684. Its clean DCEC status and RRA response have not been verified. The 1,330-image clean DC roster

is fixed: Digital-1 153, Digital-2 135, Digital-3 786, FaceDancer 107 and TextDiffuserFT 149. A run-progress snapshot records 416 completed before a later execution phase, but contains only scheduling and image properties, not pairs of final attacked E /RRA and scores. It is not a random sampled scientific outcome or a denominator for a success rate. Full multi-step result validation remains outstanding.

Table E.2: TruFor attack evaluation to insert after verifying final outputs. “Pending” is an explicit missing result, not a zero. Clean DC counts do not imply DCEC eligibility.

Family	DC n	Final evaluated n	Forged retained	Mean clean $\rightarrow$ adv E
Digital-1	153	Pending	Pending	Pending
Digital-2	135	Pending	Pending	Pending
Digital-3	786	Pending	Pending	Pending
FaceDancer	107	Pending	Pending	Pending
TextDiffuserFT	149	Pending	Pending	Pending
Total	1,330	Pending	Pending	Pending

This table needs an accompanying paired RRA distribution, zero/invalid-map count, uncompleted or excluded-image register, visual examples, and the separately calibrated DCEC transitions before it can answer the principal question. If any attack calculation fails, keep the image in the original eligible denominator and identify the failure separately. A missing gradient or time-out should not be reported as resistance.

Table E.3: TruFor attack risks and present controls. No full-population effect is claimed.

Risk	Mitigation and available evidence	Residual limitation
High-resolution gradient failure looks like resistance	Verified full-size backward route and numerical parity on manageable inputs; independent reference model reports outcomes.	Full-graph parity cannot be measured on the largest image where the original graph runs out of memory.
Pilot mistaken for a family-level result	Identify one image and its exact E /score change; keep all 1,330 DC counts visible.	Neither its DCEC membership nor population attack rate is known.
Execution progress biases the sample	Fixed DC roster and frozen ten-step settings; scheduling snapshot treated as operations only.	Final individual outcomes, failures and stem intervals still need verification.
Decision score and localisation map confused	Hard fixed score threshold for preservation; native map E is the loss.	RRA, invalid maps and final family-level score distributions remain unreported.
Metric geometry or clipping changes eligibility	Use annotated union consistently; audit native clipping and common content-coordinate convention before gate.	Review of 108 clipped rectangles, TruFor RRA and DCEC calibration are pending.
Attack works only on floats	Explicitly report continuous input, physical/model scaling and no export/re-JPEG.	Survival through common image export, compression or a service pipeline is unknown.

E.4 Final outcome table, uncertainty and visual selection

Table E.4: Required final DC→DCEC→DCEW reporting, once clean gate and attacked RRA are evaluated. Complete a separate row for each pipeline, family and split; do not fill from mean E alone.

Population	Forged	DC	DCEC	Adversarial outcomes within clean DCEC
ResNet / each selected family	Ch. 4	Ch. 4	Pending	Retain DCEC: pending; DCEW: pending; flip: pending; invalid: pending.
TruFor / each family	Ch. 4	Ch. 4	Pending	Retain DCEC: pending; DCEW: pending; flip: pending; invalid: pending.
Both eligible / each family	Ch. 4	Both DC	Pending	Report each pipeline’s outcomes separately on the same clean documents.

The completed ResNet DC-wide relative E loss already has document-stem intervals, reported in Chapter 5. For the remaining outcomes, uncertainty should also resample document stems rather than individual captures. In each resample, keep all included images from sampled stems together; recompute the paired RRA change, the DCEW numerator and the original clean DCEC denominator. Give the interval for each family’s conditional transition as well as $N_{\text{DCEC}}/n_{\text{forged}}$. The interpretation is conditional on the *frozen* gate; repeat the clean-only threshold calibration under stem resampling separately if estimating uncertainty in gate selection. Do not present an RRA or DCEW interval until those calculations exist. The one model seed, model training choices and uncertain annotation edges are additional sources of variation outside a stem-only interval.

For the main chapter, pick paired visual examples by an announced rule *before* choosing attractive panels. From the fixed DCEC set, identify per family an image near the median individual E loss, one near the lower tail (including a failed attack, if present), one with a large loss and preserved decision, and one close to the final clean or adversarial RRA boundary. Show face/text component panels for at least one Digital-2 case. Keep one DC but non-DCEC image as a coverage counterexample. In the shared-image view, show the same document for both pipelines with their own clean and adversarial maps. Every caption should include split, family, clean/attacked E and RRA, forged decisions, the rectangle convention and whether it belongs to DCEC. These examples explain mechanisms and exceptions; they cannot estimate an attack rate. The available 450 ResNet paired overlays are candidates for selection, not human ratings.

E.5 Design chronology and remaining interpretation risk

The 1/255 ResNet classification budget followed pilots, and the decision-preserving ResNet attack followed the observation that the classification control often erased its positive CAM. The full ResNet attack and a TruFor feasibility image preceded the final joint clean evidence-gate design. This sequence does not invalidate the completed paired DC E changes, but it prevents a claim that every analysis decision was preregistered before all attack outcomes were known. The future clean threshold exercise should mask the attack outcomes during the actual visual coding and gate fitting, log prior analyst exposure honestly, then freeze its values before DCEW reporting. No threshold should be adjusted to secure a particular DCEW rate.

Differences between the pipelines include ResNet training on these forgery families, TruFor’s pretrained zero-shot weights, classification calibration, map meaning and resolution, and the size of the clean-correct set. The shared-image intersection reduces the difference in images but does not remove the other differences. Accordingly, report each pipeline’s relative change from its own clean accepted evidence, show original-family coverage, and avoid treating a raw between-model percentage as an intrinsic architecture ranking. Reproducibility items such as exact scripts, runs and content digests belong to the separately reserved Appendix D, not to this scientific risk register.

Appendix F

Ethics and Security Documentation

This appendix records the ethics, research-practice, information-security and data-governance documentation supporting the study. The experimental work was conducted on research models and benchmark identity-document imagery rather than against live identity-verification services. The adversarial methods were used to evaluate the robustness of detection and localisation outputs within the defined research setting, not to demonstrate an operational verification bypass.

The FantasyID dataset has two parts that are distributed through two separate Zenodo records with different access conditions. This dissertation used only the openly accessible FantasyID dataset.

1. The openly accessible *FantasyID* dataset is published as Zenodo record 17063366 (DOI: 10.34777/c966-nn94).
2. A separate component, *FantasyID-val*, is published as Zenodo record 17063494 (DOI: 10.34777/vr5x-kb42); access to its files is restricted and subject to an End-User License Agreement (EULA).

The restricted FantasyID-val files were not acquired, accessed or used in any experiment or analysis. The relevant University ethics confirmation and required research-practice and information-security training evidence are reproduced below.

F.1 Ethics Approval

The project was submitted through the University ethics process. Figure F.1 reproduces the ethics approval or confirmation issued for the study.

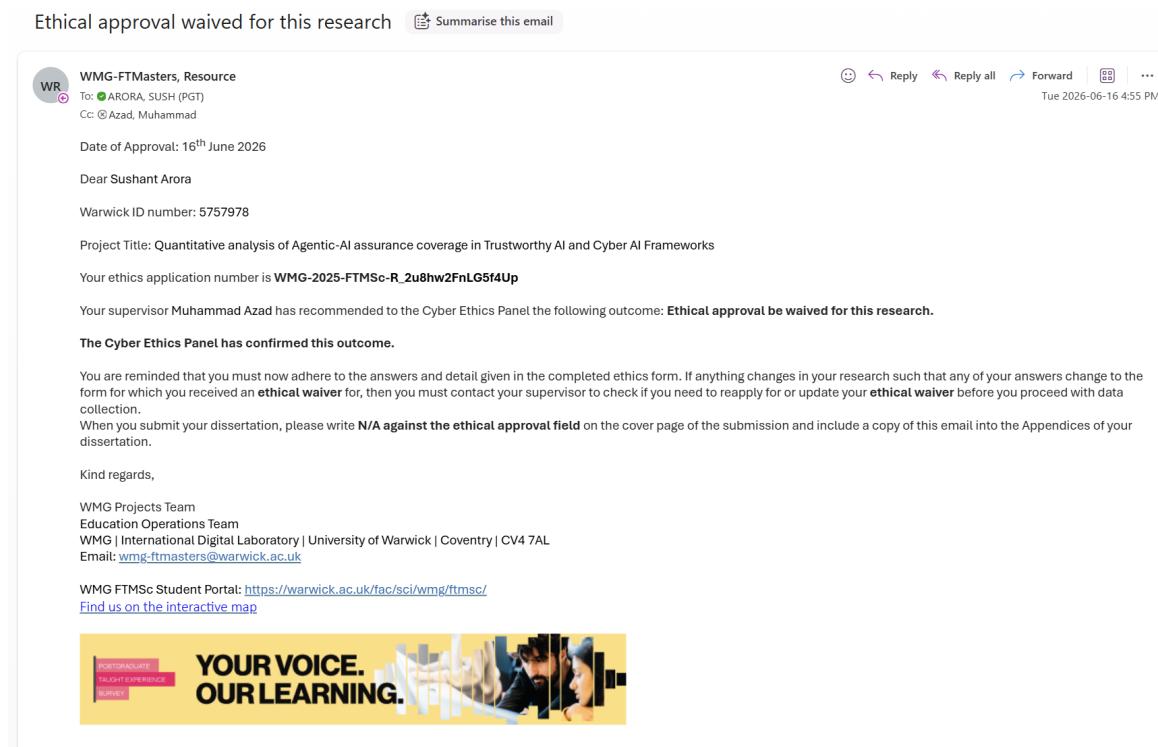


Figure F.1: Ethics approval or confirmation for the research project.

F.2 WMG Student Ethics Training

The WMG Student Ethics Training was completed as part of the research-governance requirements for the project. Figure F.2 shows the completion evidence.

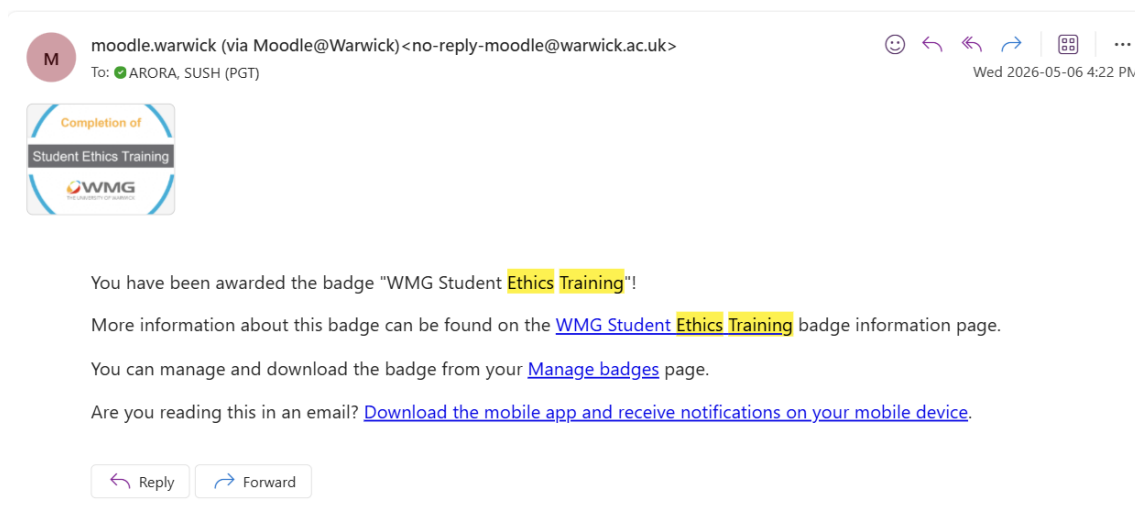


Figure F.2: WMG Student Ethics Training completion evidence.

F.3 Information Security Smart

The Information Security Smart course was completed to support appropriate handling of research data, code and computing resources. Figure F.3 shows the completion evidence.

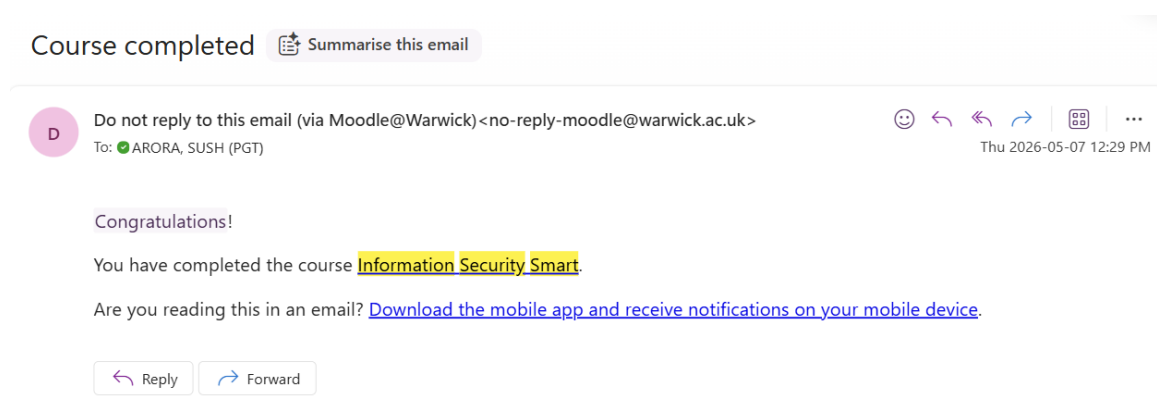


Figure F.3: Information Security Smart course completion evidence.

F.4 Responsible Research Practice

The Responsible Research Practice course was completed as part of the research training supporting this dissertation. Figure F.4 reproduces the completion evidence.



Figure F.4: Responsible Research Practice completion evidence.

F.5 Irresponsible Research Practice

The Irresponsible Research Practice course was also completed as part of the research training supporting this dissertation. Figure F.5 reproduces the completion evidence.



Figure F.5: Irresponsible Research Practice completion evidence.

Appendix G

Generative AI Declaration

G.1 Statement of Disclosure

Generative Artificial Intelligence (GenAI) tools were used as supportive tools during the planning, development and preparation of this dissertation. Their use did not replace independent academic judgement, validation of experimental outputs or responsibility for the submitted work.

GenAI was used to generate boilerplate code and generic implementation patterns. This material was not adopted unchanged: it was reviewed, tested and modified for the specific data structures, model pipelines, evaluation logic and reproducibility requirements of the project. GenAI was also used for idea refinement and as a planning aid when developing the experimental execution strategy. This included considering experiment ordering, parallel execution, checkpointing and other practical choices intended to make efficient use of the available GPU hours.